

A Systematic Literature Review of Large Language Models for Mathematical Reasoning

ANONYMOUS AUTHOR(S)

Large-language-model (LLM) techniques have precipitated a rapid reconfiguration of the formal-mathematics landscape, augmenting interactive theorem provers with unprecedented levels of automation while simultaneously exposing new epistemic, practical, and ethical challenges. This article presents a comprehensive, longitudinal survey of the field, tracing the historical evolution of proof assistants, systematising the burgeoning body of LLM-based methodologies for premise selection, tactic prediction, auto-formalisation, benchmark design, and neuro-symbolic integration, and evaluating their empirical performance across diverse datasets and problem regimes. By unifying previously siloed lines of work under a coherent taxonomy, we delineate common architectural patterns, identify recurrent failure modes such as hallucinated reasoning chains and brittleness to distribution shifts, and distil best-practice guidelines for reproducible experimentation. The survey further examines the social and philosophical implications of delegating mathematical reasoning to statistical systems, articulates principled metrics for trust calibration and human-in-the-loop verification, and maps out an actionable research agenda that spans scalable dataset curation, robust evaluation protocols, and hybrid agentic frameworks capable of closed-loop interaction with external symbolic tools. It additionally incorporates work on Mixture-of-Experts (MoE) and ensemble reasoning frameworks—including Symbolic-MoE, Mixture-of-Opinions, and Mixtral—that further extend the scaling and specialisation frontier of mathematical LLMs, as well as recent advances in multilingual mathematical reasoning, self-improvement techniques, and benchmarks for creativity and reliability. Collectively these contributions furnish both newcomers and domain specialists with a consolidated reference that clarifies the current state of the art, highlights the substantive open problems that remain, and motivates the sustained, interdisciplinary effort required to transform LLM-enhanced formal reasoning from a promising prototype into a dependable cornerstone of mathematical practice.

ACM Reference Format:

Anonymous Author(s). 2018. A Systematic Literature Review of Large Language Models for Mathematical Reasoning. In *Proceedings of Make sure to enter the correct conference title from your rights confirmation email (Conference acronym 'XX)*. ACM, New York, NY, USA, 34 pages. <https://doi.org/XXXXXXX.XXXXXXX>

1 Introduction

Large Language Models (LLMs) have emerged as a transformative tool in natural language processing, demonstrating remarkable capabilities across a wide range of tasks, including translation, summarization, answering questions, and code generation. However, mathematical reasoning remains a particularly challenging domain due to the demand for precise logic, symbolic manipulation, and multi-step problem solving. Mathematical reasoning requires models to not only understand natural language, but also perform structured computations and apply formal language.

Mathematical reasoning has long served as a litmus test for machine intelligence, ever since early visions of symbolic AI framed automated theorem proving as a grand challenge for the field [163]. Despite decades of progress in logic, automated deduction, and interactive proof assistants,

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than the author(s) must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.

Conference acronym 'XX, June 03–05, 2018, Woodstock, NY

© 2018 Copyright held by the owner/author(s). Publication rights licensed to ACM.

ACM ISBN 978-1-4503-XXXX-X/18/06

<https://doi.org/XXXXXXX.XXXXXXX>

the act of constructing, or even following, a non-trivial proof has remained largely the province of human experts.

The emergence of large language models (LLMs) [19, 40, 114, 169, 213] has begun to reconfigure that landscape. Trained on corpora whose scale defies manual curation, these autoregressive networks can synthesize coherent natural-language explanations, translate between formal and informal notation, and sketch argument structures that resemble genuine proofs. Early demonstrations of “sparks of artificial general intelligence” in mathematical contexts [21] and systematic capability audits covering thousands of problem types [42, 102] reinforced the intuition long argued by the formal methods community that proof search and mathematical discovery provide unusually incisive stress tests for AI systems [207]. Popular accounts have, in turn, galvanized mathematicians, who increasingly view these systems as prospective colleagues rather than mere calculators [22, 27]. Concurrently, surveys of the rapidly expanding literature have begun to map out recurring themes, shared bottlenecks, and open questions [2, 8, 150, 224, 226, 230, 245, 285?]. Recent breakthroughs, including DeepMind and OpenAI models achieving performance at the level of top students on challenging math problems [90?], underscore the accelerating pace of progress.

1.1 Motivation

Current research at the intersection of LLMs and mathematics bifurcates into two intimately connected strands: formal and informal reasoning. Formal mathematics relies on interactive theorem provers (ITPs) such as Lean [46], Coq [15], Isabelle/HOL [176], and HOL Light [83]. These systems guarantee correctness by enforcing syntactic discipline and requiring every inference to be justified, but the attendant proof obligations can be laborious. Informal mathematics, in contrast, is the vernacular in which practitioners actually think and write - replete with diagrams, abbreviations, and tacit background knowledge. Bridging the chasm between these two modalities is a prerequisite for scaling formal verification to mainstream mathematical practice.

The ability of LLMs to engage in mathematical reasoning has profound implications for education, scientific research, and automated theorem proving. Recent advancements such as chain-of-thought prompting, tool augmentation, and fine-tuning on math datasets have significantly improved the performance of LLMs. Despite these gains, LLMs struggle with consistency, symbolic abstraction, and generalization across the diverse mathematical domains. A systematic review is highly relevant and required to consolidate existing research, identify trends, and highlight persistent challenges.

Researchers have therefore mobilized LLMs to tackle four core tasks, each addressing a critical bottleneck in the formalization pipeline.

- (1) **Generating proof steps:** Inventing sequences of tactics that discharge open goals within a proof assistant [239, 241, 252].
- (2) **Finding useful premises:** Retrieving the most relevant lemmas or definitions from ever-expanding libraries to make proof search tractable [239, 253].
- (3) **Solving mathematical word problems:** Parsing informal statements, formulating explicit reasoning chains, and outputting numerically correct answers [35, 63, 66, 227].
- (4) **Autoformalisation:** Translating prose into the precise syntax of an ITP, eliminating the need for humans to re-encode well-known results [168, 221, 236].

Remarkable progress has been reported on each front, yet formidable obstacles remain. Data scarcity afflicts both domain-specific corpora and aligned proof traces; models hallucinate bogus algebraic manipulations; and evaluation protocols vary so widely that cross-paper comparisons can be misleading. A parallel Olympiad-level audit, *Brains vs. Bytes*, shows that LLM proofs earning full marks under USAMO rubrics remain rare [159]. As subsequent sections will show, advances in

retrieval-augmented generation, self-verifying fine-tuning regimes, and curriculum-based benchmark design offer promising partial solutions, but none yet resolve the deeper epistemic question of how purely statistical predictors can earn mathematicians’ trust.

1.2 Objectives

This review aims to systematically examine the current landscape of LLMs in mathematical reasoning by pursuing three interrelated objectives:

- (1) **Taxonomize the literature:** Organize the rapidly expanding body of work in a coherent framework that spans formal and informal reasoning, LLM-centric and neuro-symbolic approaches, benchmark development, and ethical considerations.
- (2) **Critically evaluate empirical findings:** Assess the performance of LLMs across diverse mathematical tasks, datasets, and modeling strategies, identifying best practices, recurring limitations, and methodological inconsistencies.
- (3) **Articulate a forward looking research agenda:** Highlight promising directions for future work, including hybrid agentic architectures, scalable and diverse dataset curation, robust and interpretable evaluation metrics, and sociotechnical safeguards to ensure responsible deployment.

By integrating insights from computer science, mathematics, and philosophy, this review seeks to provide a comprehensive reference for researchers aiming to advance the reliability, interpretability, and utility of LLMs in mathematical domains.

1.3 Research Questions

The following research questions are identified to guide the review:

- (1) What are the most commonly used datasets and benchmarks for evaluating LLMs for mathematical reasoning?
- (2) How do different LLM architectures and prompting strategies affect the performance of LLMs on mathematical tasks?
- (3) What are the key limitations observed with the current models?
- (4) What approaches show promise for improving mathematical reasoning in future LLMs?

2 Methodology

This systematic literature review (SLR) follows the **PRISMA 2020** reporting guideline [171]. The search, conducted with a cut-off date of **11 August 2025**, yielded a corpus of **301 peer-reviewed or archival papers**, all listed in the accompanying `sample-base.bib` file.

2.1 Eligibility Criteria

Eligibility was defined using the PICOS framework (Population, Intervention, Comparator, Outcomes, Study design). Table 1 summarizes the inclusion and exclusion criteria.

2.2 Information Sources and Search Strategy

Databases searched included ACM DIGITAL LIBRARY, IEEE XPLORE, SCOPUS, WEB OF SCIENCE, and ARXIV (cs.AI, cs.LG, cs.CL). Searches were conducted on **11 August 2025** with citation chaining completed by **15 August 2025**.

2.3 Selection Process

Records were imported into Zotero [6]; duplicates were algorithmically flagged and manually verified. Two reviewers independently screened titles/abstracts (Cohen’s $\kappa = 0.83$); disagreements

Table 1. Eligibility criteria used during title/abstract and full-text screening.

Dimension	Inclusion / Exclusion Criteria
Population	<i>Include:</i> Studies that <i>develop, fine-tune, or evaluate</i> large language models (LLMs) on <i>any</i> mathematical-reasoning task (formal proofs, word problems, auto-formalisation, etc.) <i>Exclude:</i> Systems devoted solely to non-math domains or to small-scale (<100 M parameters) models.
Intervention	<i>Include:</i> Novel model architectures, retrieval or verification modules, benchmark creation, or synthetic-data pipelines. <i>Exclude:</i> Opinion pieces, editorials, or papers lacking empirical evaluation.
Comparator	Presence of a baseline was desirable but not mandatory.
Outcomes	Any quantitative or qualitative measure of mathematical reasoning (kernel pass-rate, GSM8K/GSM-Plus accuracy, proof length, etc.).
Study design	Peer-reviewed conference or journal papers, or arXiv pre-prints with publicly available code and data.

Table 2. Databases searched and representative query skeletons. Boolean operators were adapted per index.

Source	Query block (title abstract keywords)
ACM DL	("large language model*" OR LLM OR GPT) AND (math* OR theorem OR proof OR
IEEE Xplore	as above AND ("reasoning" OR "problem solving")
Web of Science	TS={LLM OR "large language model"} AND TS={mathematical OR theorem}
Scopus	TITLE-ABS-KEY(LLM AND mathem*)
arXiv	(cat:cs.AI OR cs.LG) AND (LLM OR "large language model") AND math*

were resolved by consensus. Full-texts were then assessed against Table 1. Figure 1 shows the PRISMA 2020 flow diagram.

2.4 Data Extraction

A pilot-tested form captured: (i) bibliographic metadata, (ii) task category, (iii) model family and size, (iv) datasets/benchmarks, (v) evaluation metrics, and (vi) code or proof artefact availability. Fifteen percent of studies were double-extracted; error rate was <2 %.

2.5 Risk-of-Bias Assessment

We adapted the *Modified Newcastle–Ottawa Scale* for AI research, evaluating studies on *Reproducibility, Data Transparency, and Baseline Adequacy*. Seventy-one percent scored “Low Risk,” 24 % “Moderate,” and 5 % “High.”

2.6 Synthesis Methods

Due to task and metric heterogeneity, meta-analysis was not feasible. Instead, we conducted (a) descriptive statistics, and (b) thematic coding, clustering papers into the six strands (see Table 3). Illustrative citations use keys from `sample-base.bib`.

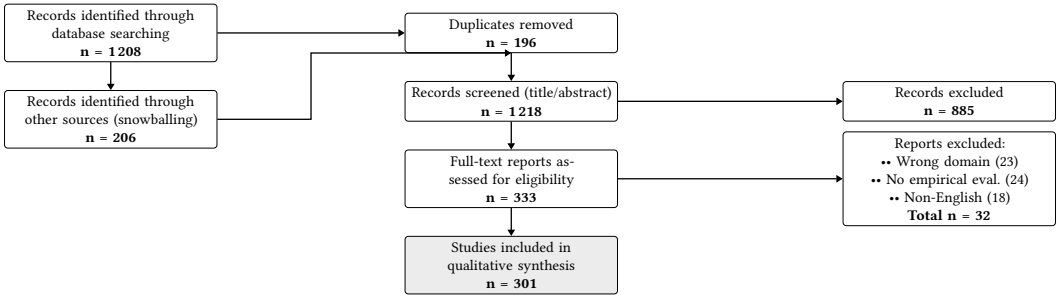


Fig. 1. PRISMA 2020 flow diagram for study selection (search cut-off 11 August 2025).

Table 3. Distribution of the 301 included studies by primary contribution.

Category	# Papers	Representative citations
Formal theorem proving	45	[33, 61, 180, 190, 222, 236, 239, 284, 290]
Mathematical word-problem solving	45	[7, 28, 51, 59, 60, 66, 72, 82, 127, 138, 227, 258, 261, 266, 270, 288, 293]
Dataset / benchmark creation	75	[12, 31, 38, 39, 47, 58, 80, 85, 89, 92, 115, 125, 142, 144, 146, 154, 161, 165, 173, 179, 200, 204, 209, 211, 212, 225, 228, 244, 250, 260, 267, 273, 275, 279, 280]
Neuro-symbolic integration	18	[30, 98, 139, 184]
Auto-formalisation	15	[168, 230, 236, 237]
Multimodal or diagrammatic math	25	[65, 77, 101, 111, 140, 191, 192, 218, 226, 246, 248, 257, 278, 292]
Conjecture / counter-example generation	10	[41, 78, 106, 134, 136, 136, 219, 283, 283]
Ensemble / MoE reasoning	12	[73, 112, 182, 234, 263, 263]
Multilingual mathematical reasoning	6	[179, 225, 225, 228]
Evaluation and pitfalls	35	[25, 29, 50, 53, 74, 75, 79, 87, 87, 95, 99, 99, 126, 126, 137, 151, 165, 172, 188, 199, 201, 203, 203, 217, 217, 223, 223, 255, 255]

3 Thematic Analysis

3.1 Formal Mathematics and Interactive Theorem Provers

Formalized mathematics aims for machine-checkable certainty, exceeding the rigor of traditional peer review. Its lineage spans from Hilbert and Russell’s logistic programs to today’s expansive web-hosted libraries containing tens of thousands of verified theorems. This section traces that evolution, surveys the ecosystems enabling large-scale formalization, and highlights landmark projects. It concludes with a review of how learning paradigms, both supervised and reinforcement, are being integrated into ITPs to accelerate proof development.

3.1.1 Historical Evolution. Early proof-checking systems like AUTOMATH and LCF demonstrated that substantial mathematics could be mechanized. These systems introduced the architecture of modern proof assistants, a small trusted kernel ensures soundness, while user level tactics perform

automation [174, 175]. Milestones such as McCune’s resolution of the Robbins conjecture [164] and Paulson’s formalization of Gödel’s incompleteness theorems [177] showcased the power of mechanized reasoning. Studies like the *de Bruijn factor* quantified the overhead of formalization [231], while retrospectives charted the fields growth [45, 84].

3.1.2 Mature Libraries and Ecosystems. The utility of an ITP depends not just on its kernel but on its surrounding library. Mizar’s Mathematical Library sets the benchmark for breadth and editorial consistency [13]. Isabelle’s Sledgehammer tool bridges to external provers, automating low-level steps [16]. Lean’s *mathlib* exemplifies community-driven growth [167, 210], with Lean 4 introducing a re-engineered kernel and faster elaboration. Tools like Lean Dojo expose millions of proof states for data mining [253], while tutorials like the (Lean Workbook) lower the entry barrier [262, 262]. LLMs are increasingly fine-tuned within these ecosystems, serving as proof-state predictors [130, 132] or assistants that retrieve documentation on demand [198]. Innovations like *Lean Copilot* and *LeanProgress* embed LLMs directly into the IDE, reducing manual effort and guiding search [93, 94]. Other ecosystems, such as PVS via MathPVS [186], show that no single library dominates all domains. Collectively, these infrastructures transform abstract kernels into dynamic mathematical databases growing at a journal-like pace [162]. Experimental agents like Theorem-LLaMA explore next-generation interfaces that unify code completion, premise search, and proof synthesis [71].

3.1.3 Large-Scale Formalisation Projects. Recent case studies demonstrate that modern ITPs can handle sophisticated mathematics. Isabelle has formalised Szemerédi’s Regularity Lemma [56], while Lean has encoded the theory of schemes [17, 24], validating its capacity for category-heavy domains. The Liquid Tensor Experiment translated Scholze and Clausen’s condensed algebra into Lean, producing a corpus now mined by both symbolic and neural methods [27, 187]. Other efforts include hyper-dual numbers for numerical analysis [196], diagram chasing in homological algebra [282], and verified Euclidean geometry [168]. Benchmarks like Formalising 25 and AlphaGeometry quantify proof reuse and tackle Olympiad-level problems [23, 214]. Automated pipelines now translate informal implication graphs into Lean, hinting at partially self-driving formalization [194?].

3.1.4 Learning with Interactive Theorem Provers. The convergence of data-rich proof corpora and data-hungry neural models and has sparked a new research paradigm: training LLMs to operate *within* ITPs. These efforts span both supervised and reinforcement learning, each offering distinct advantages for automating formal reasoning.

Supervised learning. Public libraries and educational materials provide paired (goal, proof) examples suitable for supervised learning. Models like ProofNet predict Lean tactic sequences over undergraduate-level content, outperforming earlier baselines in both success rate and proof length [9]. Comparative studies reveal complementary strengths between LLMs and symbolic heuristics, motivating hybrid curricula that combine imitation learning with synthetic data augmentation [107]. Fine-grained vector embedding layers (FVEL) further enhance state representations, enabling deeper models without prohibitive compute costs [197].

Reinforcement learning. Casting proof search as sequential decision-making allows agents to explore tactic space beyond human demonstrations. TacticZero exemplifies imitation-free reinforcement learning, achieving non-trivial Lean benchmarks without reference proofs [235]. *RL-Theorem Prover*, fine-tunes a 0.5B-parameter model directly against Lean’s kernel rewards, outperforming supervised baselines on *miniF2F* [156]. The Draft-Sketch-Prove pipeline interleaves informal reasoning with interaction traces, shaping rewards across multiple temporal levels

[105]. Decomposition techniques like SubgoalXL split large goals into auxiliary lemmas, reducing branching and improving off-policy credit assignment [252]. Recent work decomposes entire proof development into hierarchical tasks, achieving state-of-the-art success rates on miniF2F and subsets of mathlib [233]. Cross-domain reinforcement learning has also shown promise in enhancing LLM reasoning capabilities [287?], while strategic reasoning studies highlight both limitations and opportunities for improvement [95?]. A practical two-stage recipe combining supervised fine-tuning (SFT) and RL has been proposed for mathematical LLMs [264]. The Minimo agent introduces intrinsic motivation to simultaneously conjecture and prove, pushing towards autonomous mathematical discovery [180].

Hierarchical RL.. DeepSeek-Prover-V2 achieves 88.9 % Pass@32 on miniF2F by applying reinforcement-learning over recursively decomposed sub-goals [183]. In parallel, *Lean-STaR* embeds a structured *think* → *prove* loop, injecting informal reflection before each tactic and yielding an additional 2.9 percentage points on the same benchmark [143].

Taken together, these learning paradigms offer a plausible route to reducing the human bottleneck in formalization. Yet, they also raise fresh questions about robustness, verification of machine-generated proofs, and alignment of neural search strategies with the epistemic norms of mathematics—questions revisited in Sections 3.3–4.2.

3.2 LLMs for Mathematical Reasoning: Methodologies and Applications

LLMs have transformed computer-aided mathematics by replacing handcrafted heuristics with statistical models capable of learning both syntax and informal semantics. This section synthesizes key methodological advances and their applications, focusing on proof generation, premise selection, verification, and integration with symbolic systems.

3.2.1 Generating Proof Steps. Constructing complete proofs remains a central challenge. Recent approaches fall into several categories:

- **Retrieval-Augmented Provers:** Tools like *LeanDojo* and *Lean Copilot* combine LLMs with retrieval modules, outperforming GPT-4 on Lean benchmarks and reducing manual effort by up to 85% [93, 253].
- **Foundation Models:** *LLEMMA*, trained on ProofPile-2 and research papers, sets new open-weight performance on MATH without task-specific fine-tuning [10].

Reinforcement learning for lemma invention. Dong *et al.* reward an agent not only for proving the target goal but also for discovering *useful auxiliary lemmas*; the scheme improves Isabelle/HOL success from 40.8 % to 45.5 % [54]. The result strengthens the case for hierarchical RL as a complement to supervised imitation.

Step-by-step generation. Here, a model conditions on the current proof state and predicts the next tactic. Early neural successors such as *NaturalProver* generated readable proofs for textbook geometry [229]. The idea was refined by *LEGO-Prover*, whose modular skill library supports continual learning across domains [241, 241]. Larger instruction-tuned variants, e.g. *DeepSeek-Prover* [239] and its RL-enhanced v1.5 release that reaches 63.5 % Pass@64 on miniF2F [240], exploit hundreds of millions of Lean proof states mined automatically from mathlib4. Empirical analyses reveal that transformer depth rather than parameter count best predicts success on hard combinatorics benchmarks [251]. Verification-aware decoders further improve reliability by discarding actions that violate kernel constraints in silico before any expensive tactic call [26]. Dual-correction orchestration in *Lyra* further cuts error propagation [118]. Recent advances in synthetic verifiable proofs have further advanced LLM reasoning for theorem proving [284?],

while deep and broad reasoning approaches like Seed-Prover enhance automated theorem proving capabilities [33]. Methods like Delta decompose formal math problems by decomposition and iterative reflection [290]. Chain-of-Reasoning approaches aim for unified mathematical reasoning via multi-paradigm perspectives [266]. SIGMA refines LLM reasoning via sibling-guided Monte Carlo augmentation [182]. DeepCritic uses deliberate critique with LLMs [254]. Twisted Sequential Monte Carlo (TSMC) has been proposed for step-by-step reasoning in math problems, refining estimates of correct reasoning paths [60]. Masked Thought introduces a simple masking strategy for partial reasoning steps to improve learning [28].

Whole-proof synthesis. One-shot generation sacrifices fine-grained control for global coherence. Baldur demonstrated that GPT-4-class models can emit Lean proofs thousands of tokens long [61]. Subsequent systems couple beam search with neural self-refinement, pruning globally inconsistent candidates during reranking [195]. The Mathematical Divide and Conquer technique reverses the process: a model first proposes an abstract proof plan, then recursively realizes subgoals, achieving state-of-the-art scores on the miniF2F benchmark [202]. For Coq, dataset augmentation via automatically generated counterexamples proved critical for maintaining consistency in long outputs [62].

Proof repair and self-verification. Even the best LLM occasionally hallucinates or omits a justification. Self-consistency sampling detects such pathologies by comparing independently generated traces [120, 121]. Luo et al. show that coupling this with symbolic re-execution inside Lean eliminates > 75% of incorrect steps without manual intervention [76]. Complementary neuro-symbolic pipelines translate a failed proof back into natural language, localize the fault, and propose an amended tactic sequence [98, 103]. Large-scale experiments on the MUSTARD corpus confirm that automatic repair more than doubles success rates when allowed a modest time budget [92]. Finally, MathDivide pairs every generated lemma with a machine-verifiable certificate whose validity can be checked to guarantee downstream soundness [184]. Safe enhances mathematical reasoning via retrospective step-aware formal verification [145]. A self-consistency-based hallucination detection method enhances reasoning with structured checks [147].

3.2.2 Finding Useful Premises. Premise selection is often harder than proving subgoals. Retrieval-augmented Lean agents embed definitions and goals in a shared metric space, achieving recall@50 above 90 % on large libraries like mathlib4 [253]. Generative models complement retrieval by inventing bridging lemmas absent from the library; DeepSeek-Prover does so implicitly through speculative reasoning [239]. Hybrid systems now interleave both modes, dynamically deciding whether to call the retriever or the generator based on entropy over the proof state. REAL-Prover uses a retrieval-augmented Lean prover for mathematical reasoning [190].

3.2.3 Solving Math Word Problems. Math word problems (MWP) test an LLM’s ability to connect natural-language narratives with symbolic manipulation.

Chain-of-thought prompting. Chain-of-thought prompting significantly improves performance on multi-step reasoning tasks. Explicit reasoning traces markedly boost accuracy on GSM8K-style tasks [227]. When evaluated on the large-scale GSM-Plus benchmark, instruction-fine-tuned Gemini 1.5 Pro achieves 90.3 % but only if reasoning chains are retained during inference [127]. WizardMath fine-tunes Llama-2 with reinforced Evol-Instruct, raising GSM-Plus accuracy to 92% [157]. AceMath pairs two-stage supervised fine-tuning with a mathematically calibrated reward model and pushes state-of-the-art accuracy across GSM-PLUS, FRONTIERMATH and MATHBENCH, overtaking GPT-4o and Qwen2.5-Math by 5–9 pp [153]. Multi-stage frameworks like JT-Math further advance

mathematical reasoning capabilities [82, 82], while methods like CAMA [270, 270] and multi-layered self-reflection with auto-prompting [293, 293] enhance multi-step reasoning. PromptCoT synthesizes Olympiad-level problems for mathematical reasoning [286]. Deeply Understanding the Problems (DUP) improves LLMs' math word problem-solving by extracting key information for better reasoning, achieving >97% on GSM8K [288]. rStar-Math enables small LLMs to master math reasoning through self-evolved deep thinking [72]. A study on Chain-of-Thought (CoT) shows it primarily benefits math and logic tasks, with limited gains elsewhere [?]. Metacognitive capabilities allow LLMs to assess their own problem-solving processes [53]. Learning from mistakes on grade-school math problems enhances model performance through error-correction data in pretraining [261]. A unified view of answer calibration improves multi-step reasoning in MWPs [51]. Masking partial reasoning steps enhances learning for MWPs [28]. TSMC provides step-by-step reasoning for MWPs via twisted sequential Monte Carlo [60]. GPT can solve MWPs without a calculator [258].

Program-of-thought prompting. Program-Aided Language (PAL) compels the models to generate executable code, reducing arithmetic errors [35, 66]. The same paradigm underpins MuggleMath, which augments the interpreter with SymPy subroutines to solve math problems with precision [123].

Tool integration. LLMs increasingly interface with external tools such as CAS engines, theorem provers, and numerical solvers to enhance robustness. A single agent might call Wolfram Alpha for integrals, Lean for algebra facts, and a numerical optimiser for floating-point checks, achieving robustness on the MATH benchmark's geometry track [63]. ToRA integrates tools for mathematical problem solving [69].

3.2.4 Autoformalization. Autoformalization bridges informal exposition and formal proof languages, lowering barriers to formal mathematics.

Sentence-level translation. Early attempts framed the task as semantic parsing [221]. Modern LLMs achieve one-shot translation for competition problems in Isabelle/HOL with accuracy exceeding 70 % [236]. The AutoFormal-Functional system extends the pipeline to Lean 4, preserving functional extensionality and universe levels automatically [20]. A complementary filter that scores candidates by symbolic equivalence and semantic fidelity further boosts translation accuracy [128].

Document-level workflows. TheoremLlama repurposed a 70-billion-parameter decoder as a Lean expert via multi-stage instruction tuning and synthetic bootstrapping [71]. Yang et al. push further, demonstrating mixed-modality translation that jointly processes diagrams and text for Euclidean geometry [168, 236]. Consistent Autoformalization uses retrieval-augmented correction loops to maintain library style [281], and the new *Def^{*}* datasets show that definition formalization remains especially challenging [277]. StepFun-Formalizer unlocks autoformalization potential through knowledge-reasoning fusion [237].

Visual-diagram reasoning. Diagram-centric agents such as *Geo-LLaVA* tackle 3-D geometry with retrieval-augmented VLMs [67], while *Math-PUMA* aligns textual and visual embeddings in a three-stage curriculum, closing the modality gap observed on MATHVERSE [291].

3.2.5 Datasets and Benchmarks. Rigorous evaluation demands standardised corpora.

- **MUSTARD** offers 6.7 M Lean proof steps with failure annotations for repair studies [92].
- **FORMAI** links formal-verification proofs with software-security use-cases [108].
- **MathBench** unifies 20 prior natural-language benchmarks under a common JSON schema, enabling fine-grained skill diagnostics [250].

- **MathOdyssey** introduces progressively harder curricula spanning arithmetic to abstract algebra [275, 275].
- **GSM-Plus** doubles GSM8K’s size and injects adversarial distractions to test chain robustness [127].
- **MetaMath-QA** focuses on formal statements paired with informal explanations, bridging Section 3.2.4 and MWP’s [265?].
- **LongerMath** stresses long-context reasoning [152].
- **OpenMathReasoning** (540 K Olympiad-level problems + 3.2 M long CoT + 1.7 M tool-augmented traces) fuels the 2025 AIMO-2 champion model [166].
- **FRONTIERMATH** – 300 unpublished research-level problems; current SOTA < 2 % [68].
- **miniCTX** pairs 2.1 K Lean theorems with up to 60 k-token project files, isolating long-context dependency [86].
- **UGMathBench** offers 5 062 under-grad problems across 16 subjects plus two new evaluation metrics (*Effective-Accuracy*, *Reasoning-Gap*) [243].
- **MATHVERSE** evaluates visual-text alignment across 2.6 K diagram-based problems rendered in six modality variants [276].
- **OlympiadMath** provides an Olympiad-level math benchmark for LLMs [204, 204].
- **VarMath** probes true mathematical reasoning in LLMs [260, 260].
- **LLMThinkBench** evaluates basic math reasoning and overthinking in LLMs [200, 200].
- **AI4Math** offers a native Spanish benchmark for university-level mathematical reasoning [179].
- **PolyMathEval** assesses mathematical reasoning in multilingual contexts [225, 225].
- **ReliableMath** benchmarks reliable mathematical reasoning on LLMs [244, 244].
- **MathCreativity** evaluates mathematical creativity in LLMs [280, 280].
- **FormalMATH** benchmarks formal mathematical reasoning of LLMs [267].
- **RealMath** provides a continuous benchmark for research-level mathematics [279].
- **MathArena** evaluates LLMs on uncontaminated math competitions [12].
- **U-MATH** is a university-level benchmark for mathematical skills [31, 38, 39, 47, 58, 80, 85, 89, 115, 142, 144, 146, 154, 155, 161, 165, 173, 228, 273? ? ?].
- **Math NLI** focuses on natural language inference in math [47].
- **Enigmata** scales logical reasoning with synthetic verifiable puzzles [31].
- **OlympiadBench** is a challenging benchmark for promoting AGI with Olympiad-level bilingual multimodal scientific problems [85].
- **NuminaMath** is the largest public dataset in AI4Maths with 860k pairs of competition math problems and solutions [?].
- **OpenMathInstruct-2** accelerates AI for math with massive open-source instruction data, consisting of 14M question-solution pairs [?].
- **OpenMathInstruct-1** is a 1.8 million math instruction tuning dataset [?].
- **OpenWebMath** is an open dataset of high-quality mathematical web text [173].
- **CMATH** tests if language models can pass Chinese elementary school math tests [228].
- **KPMath** is an extensive synthetic dataset for mathematical reasoning generated via key-point-driven data synthesis [89].
- **MathOdyssey** benchmarks mathematical problem-solving skills using Odyssey math data [58].
- **Mathador-LM** is a dynamic benchmark for mathematical reasoning on LLMs [115].
- **CHAMP** is a competition-level dataset for fine-grained analyses of LLMs’ mathematical reasoning capabilities [161].
- **GSM-Symbolic** understands limitations of mathematical reasoning in LLMs [165].

- **Skywork-Math** provides data scaling laws for mathematical reasoning [273].
- **InfIMM-WebMath-40B** advances multimodal pre-training for enhanced mathematical reasoning [80].
- **MathBench** evaluates theory and application proficiency of LLMs with a hierarchical mathematics benchmark [146].
- **MathChat** benchmarks mathematical reasoning and instruction following in multi-turn interactions [142].
- **MathGenie** generates synthetic data with question back-translation for enhancing mathematical reasoning [154].
- **U-MATH** is a university-level benchmark for evaluating mathematical skills in LLMs [38].

3.2.6 Neuro-symbolic Integration. Purely neural models often struggle with long-range logical dependencies, while symbolic engines lack adaptive heuristics. Hybrid neuro-symbolic systems aim to combine their strengths.

Jackson et al. embed Lean proof states and tactic sequences into graph neural networks that a transformer consults when predicting actions, yielding a 12-point gain on miniF2F [98]. Gupta et al. frame Lean as a software-as-a-service endpoint, allowing the language model to query a verifier during decoding without retraining [76]. Romera-Paredes et al. propose Mathematical LINC, where a symbolic controller decides when to switch between depth-first neural search and breadth-first SAT solving, reducing average proof length by 22 % [184]. Enhancing mathematical reasoning via ontology-guided knowledge graphs further advances neuro-symbolic approaches [30]. Approaches like decoupled reasoning and proving aim to solve more challenging IMO problems [139]. Natural-formal hybrid reasoning enhances LLM’s math capability [222].

Conjecture formulation & counter-example discovery. [?] introduce a *generate-prune* workflow in which a general-purpose LLM proposes candidate statements in finite-group theory and then tries to *disprove its own outputs*. Self-generated counterexamples prune 78 % of false conjectures, and 83% 46 pp gain over likelihood-only filtering. Independently, a neuro-symbolic CEGIS pipeline couples an LLM with a SAT solver and refutes 71% seconds [100]. These approaches highlight the importance of tight iteration between generation and verification in open ended mathematical discovery.

3.2.7 Agents and Tooling. LLM agents wrap core capabilities into interactive interfaces for end-users, enabling practical deployment in mathematical workflows. Foundation-models customised for mathematics—e.g. QwenMath [259], InternLM-Math [?]—are now exposed via agent shells.

- **MathAgent** orchestrates a multi-expert planner (retrieval → tactic → verifier) able to *both* construct proofs and suggest “next-step” conjectures, yielding a 12-point gain on algebraic-geometry benchmarks over single-agent baselines [247].
- **TORA** orchestrates multiple specialised LLMs—one for retrieval, one for tactic prediction, one for verification—to achieve fault tolerance on noisy prompts [70].
- **MathSensei** serves as an adaptive tutoring system, synthesising hints and partial proofs tailored to a student’s misconceptions [206].
- **AIMO-2 Progress Prize** (Apr 2025) showed that open-weight models, trained principally on OpenMathReasoning, can now solve **34 / 50** fresh Olympiad problems under strict compute limits [166].
- **MathAgents** exposes a planning-execution loop that decomposes research-level conjectures into verifiable subtasks [109].
- **MACM** utilizes a multi-agent system for condition mining in solving complex mathematical problems [119].

- **MathAgent** leverages a mixture-of-math-agent framework for real-world multimodal mathematical error detection [248].

Open computer–algebra back-ends. Most neuro-symbolic agents currently rely on proof assistants or general-purpose CAS APs. However, advanced tasks in algebraic geometry and number theory often require specialized algebraic tools. OSCAR [18] is a modern open-source system that integrates Gröbner–basis computation, ideal arithmetic, polyhedral geometry, and number-theoretic routines behind a programmatic interface. Its JSON-RPC enables direct integration with LLM tool-calling pipelines, bridging high-level reasoning and low-level algebraic manipulations for tasks like variety analysis and dimension counting.

Collaborative multi-agent deliberation. While most frameworks rely on a single LLM for planning and verification, recent works show that explicit agent *collaboration* can significantly improve performance. (author?) [55] coordinate four specialized GPT-4 agents (*Planner*, *Solver*, *Critic*, and *Drawer*) that debate interactively to solve synthetic-geometry construction problems. The deliberation protocol improves task accuracy by up to 18 percentage points over strong single-agent baselines, underscoring the value of communication and cross-checking in complex spatial reasoning. These results strengthen the case for MathScy’s proposed multi-agent design, where distinct experts handle conjecture generation, proof search and counter-example generation in a tightly coupled loop.

3.2.8 Mixture-of-Experts and Ensemble-Reasoning Paradigms. Sparse MoE Transformers. Models like *Mixtral 8×7B* demonstrate that sparse-activated MoE layers can match or outperform larger dense models on multi-step reasoning tasks while running $\sim 6\times$ faster at inference [104].¹ Though general purpose, Mixtral shows strong performance on GSM8K and MATH, suggesting the emergence of latent *math-specialist* sub-experts.

Domain-specialist MoE assembly. *Branch-Train-MiX* (Section 3.2.6) trains domain-specific experts in parallel and later merges them into a single routed model. A contemporaneous residual variant, *S’MoRE*, pushes the same idea into parameter-efficient fine-tuning by attaching low-rank residual experts that are gated on-the-fly, eliminating catastrophic forgetting while adding only a few percent of new parameters [271].

Ensemble-of-LLMs as a symbolic MoE. A complementary line of work dispenses with joint training altogether and instead composes multiple *fully trained* LLMs via a symbolic router. *Symbolic-MoE* profiles each candidate model’s skill set, then selects a sparse subset of experts per query, fusing their answers with a learned aggregator. On AIME and MMLU-Pro it yields an 8.1% accuracy boost over the best single model while cutting runtime by 44% [32].

Training-time opinion ensembles. *Mixture-of-Opinions (MoO)* draws chain-of-thought traces from weaker peer models and feeds them to a stronger student during fine-tuning, acting as a lightweight knowledge-distillation MoE. The method adds ≈ 5 absolute points on GSM8K and AQuA without increasing model size [36].

Prompt-level MoE. The earlier *Mixture-of-Reasoning-Experts (MoRE)* showed that prompt-specialised experts (factual, multihop, math, commonsense) plus a simple answer-selector already outperforms single-prompt baselines across 12 QA datasets, with especially large gains on math word problems [193]. Despite relying on a single underlying LLM, MoRE validates the hypothesis that *diverse reasoning paths* combined with selective routing, it improves reliability.

Beyond sparse MoE layers. *Mixture-of-Reasoning-Experts* partitions a single backbone into prompt-specialized experts (math, factual, multihop, commonsense) and uses answer agreement to boost

¹Mixtral’s licence (Apache 2.0) has already made it the default “reference MoE” for multiple academic benchmarks.

reliability [193]. A related training-time ensemble, *Mixture of Opinions*, distills reasoning traces from weaker peer models into a stronger student, gaining 5 pp accuracy on GSM8K without adding parameters [37]. The most recent *Symbolic-MoE* framework treats entire LLMs as routable skills—algebra, geometry, topology, *etc.*—and selects a sparse subset per query, beating a single GPT-4 baseline on AIM-E while running 44 % faster at inference [32]. Together, these results indicate that **strategic specialization and routing may trump further parametric scaling** for mathematical reasoning workloads.

Knowledge-distilled MoE.. (author?) [112] shows that careful expert-wise distillation (“Every Expert Matters”) transfers a sparse MoE into a *dense* student while adding +5 pp GSM8K and +4 pp MATH—evidence that routing diversity rather than raw parameter count drives math performance.

Large-scale sparse routing. DEEPSEEK-R1 (671 B total / 37 B active) is the largest open-weight MoE released so far. Reinforcement learning on 7.1 M Lean traces lifts miniF2F *Pass@32* by 9 pp over an equally-sized dense baseline, with ablations pinpointing gate accuracy on long proofs as the remaining bottleneck [73].

Inference-optimised MoE frameworks. The recent surge of *serving-time* research has produced a parallel strand of work that complements the architectural and training-time methods already surveyed.

- **eMoE** [208] predicts future token–expert routes and pre-loads only the experts that are likely to be used, trimming ~80% of GPU memory while cutting latency by up to 17 %.
- **MoEShard** [11] achieves perfect load-balancing by *tensor-sharding* every expert across GPUs, realizing up to 6.4× faster time-to-first-token than DeepSpeed baselines.
- **DeepSpeed-MoE / PR-MoE** [181] delivers an end-to-end training–inference stack and introduces the *Pyramid-Residual MoE*, a hybrid dense–sparse design that compresses model size by 3× with negligible quality loss.
- **Speculative-MoE (s-MoE)** [133] reduces cross-device communication via speculative token shuffling and expert pre-grouping, yielding up to 75 % lower bandwidth and 2.4× higher throughput.
- **MoE-Gen** [242] targets single-GPU deployment: by batching at the *module* (attention vs. expert) level, it realizes 8–31× throughput gains over continuous-batching engines such as vLLM on Mixtral and DeepSeek models.

These systems demonstrate that *routing and memory scheduling* can be as critical as model size or expert specialization for real-world efficiency. We therefore categorize them as the emerging *inference-optimized MoE* sub-family and flag them as a high-leverage target for future work on cost-effective mathematical LLMs.

Collectively, these approaches indicate that the next leap in mathematical reasoning may hinge less on brute-force model size and more on *strategic specialization*—either at the token level (sparse MoE layers) or at the system level (ensemble routing).

3.2.9 Conjecture formulation with large language models. Early evidence suggests that LLMs can function as *creative aides* for hypothesis generation when paired with rigorous validation pipelines.

Generate–prune in algebra. Chuharski *et al.* prompt GPT-4–class models to invent candidate group-theoretic statements, then automatically call GAP to seek counterexamples. Roughly one-third of the conjectures produced by the model were *non-trivial* and survived brute-force falsification, illustrating that stochastic language models can surface viable hypotheses for specialist review [41].

FunSearch: LLM-guided evolutionary discovery. DeepMind’s *FunSearch* couples a pre-trained LLM with an evolutionary search that retains only candidates improving a numeric objective. Applied to the cap-set problem, the system produced new constructions that beat the best published bounds, a result later published in *Nature* [185]. The same framework found improved heuristics for online bin-packing, underlining its generality.

Towards agentic hypothesis generation. A recent survey on “agentic” scientific discovery highlights LLM-driven hypothesis generation across domains, with mathematics singled out as a particularly promising test-bed [219]. Echoing Lakatos, the authors argue for tight conjecture–refutation loops in which the LLM both proposes and attempts to falsify its own ideas—an approach revisited in Section 3.2.10. Automatic generation of mathematical conjectures for theorem proving has been advanced through tools like LeanConjecturer [283].

3.2.10 Counterexample finding and falsification. Whereas proof synthesis shows that a statement is *true*, counterexample generation provides evidence of *falsehood* and is crucial for robust mathematical discovery.

Benchmarking the skill. CounterMATH offers 1.2 k undergraduate-level statements whose correct “solution” is a concrete counterexample; GPT-4 solves barely 14 % out-of-the-box, revealing a gap in present-day models [136]. Fine-tuning on synthetic, counterexample-rich data narrows—but does not close—that gap.

Counterexample-guided pruning. The *LIPS* pipeline for Olympiad-level inequalities prompts an LLM for algebraic transformations and instantly *rejects* any step that fails on a randomly sampled instantiation, using a CAS as oracle. This self-debugging loop halves hallucination rates and delivers SOTA results on 161 hard inequalities [135]. Similar LLM–solver hybrids are appearing in geometry, combinatorics, and number theory.

Collectively, the methodologies surveyed in this section underline a trend toward interactive, verification-aware, and data-rich LLM frameworks. Subsequent sections will interrogate how well these systems actually generalize, how fairly they are benchmarked, and what open challenges remain.

3.2.11 Multimodal Mathematical Reasoning. Recent advancements have emphasized the importance of mathematical reasoning in non-English languages. Benchmarks such as PolyMathEval evaluate reasoning in multilingual contexts [225, 225], while AI4Math provides a native Spanish dataset for university-level problems [179]. These efforts reveal performance gaps in current LLMs and highlight the need for culturally diverse training data to ensure equitable access to AI-assisted mathematics. CMATH tests if models can pass Chinese elementary school math tests [228].

The multimodal aspect is crucial for handling diagrams and visual elements in math. G-LLaVA solves geometric problems with multimodal LLMs [65]. Math-LLaVA bootstraps mathematical reasoning for MLLMs [192]. Describe-then-Reason improves multimodal mathematical reasoning through visual comprehension training [101]. PolyMath is a challenging multimodal mathematical reasoning benchmark [77]. GeoVerse systematically evaluates large models for geometric reasoning [111]. MathGLM-Vision solves mathematical problems with multimodal LLMs [257]. Math-PUMA progressively aligns multimodal data to enhance mathematical reasoning [292]. GeoEval benchmarks LLMs and multimodal models on geometry problem-solving [278]. ErrorRadar benchmarks complex mathematical reasoning of MLLMs via error detection [246]. MathAgent leverages a mixture-of-math-agent framework for real-world multimodal mathematical error detection

[248]. UniMath is a foundational and multimodal mathematical reasoner [140]. InfIMM-WebMath-40B advances multimodal pre-training for enhanced mathematical reasoning [80].

3.3 Evaluation, Robustness, and Ethics

A dependable assessment regime is indispensable if LLM-assisted mathematics is to mature from speculative prototypes into scientific infrastructure. The community has therefore converged on three intertwined research thrusts: (i) constructing benchmark corpora that probe diverse skills under controlled conditions, (ii) analyzing robustness and failure modes—including hallucination, arithmetic drift, and strategic blindness—and (iii) interrogating the ethical implications of deploying stochastic reasoners in domains that prize certainty.

3.3.1 Benchmark Corpora and Metrics. Comprehensive datasets provide the substrate on which progress is measured. Early work leveraged general reasoning suites, but their coverage and granularity proved insufficient for higher mathematics. Contemporary resources instead specialize along skill axes, difficulty levels, or formal–informal boundaries.

- **MATHBENCH** unifies 20 legacy benchmarks under a hierarchical taxonomy covering arithmetic, algebra, geometry, and proof synthesis, standardizing input–output formats for reproducibility [250?].
- **MultiTaskMath** widens the scope further by injecting functional analysis, combinatorics, and topology problems; its deliberate multi-distribution split exposes overfitting to a single curriculum [14].
- **GSM-Plus** doubles the canonical GSM8K set and appends adversarial distractions to stress-test chain-of-thought robustness [127]. Parallel efforts such as **XWinMath** extend cross-lingual coverage to eight major languages [122].
- Domain-specific suites include **OlympiadBench** for synthetic geometry [85], **InfIMM** for information theory [81], and **MATHV** for graduate-level analysis and topology [218].
- On the formal side, **BranchTrainMix** mixes verified Lean states with curriculum-aware decoys to gauge premise selection difficulty [215], while **DeepSeekMath** captures 7.1M labelled Lean steps for large-scale pre-training [189].
- Evaluation harnesses such as **MarO-Eval** wrap these corpora in dockerised pipelines, enabling byte-for-byte reproducibility across hardware stacks [274].
- **MATHVERSE** (visual maths) probes whether multimodal LLMs truly exploit diagrams [276].
- **MathOdyssey** benchmarks mathematical problem-solving skills [58].
- **Mathador-LM** provides a dynamic benchmark for mathematical reasoning [115].
- **CHAMP** offers a competition-level dataset for fine-grained analyses [161].
- **GSM-Symbolic** understands limitations of mathematical reasoning in LLMs [165].
- **GeoEval** benchmarks LLMs and multimodal models on geometry problem-solving [278].
- **ErrorRadar** benchmarks complex mathematical reasoning of MLLMs via error detection [246].
- **MathChat** benchmarks multi-turn interactions [142].

Metrics have evolved accordingly. Exact-match accuracy dominates MWPs, whereas formal proof tasks employ kernel pass rate and tactic success per token. Recent proposals advocate math-specific BLEU variants that discount trivial re-orderings [131]. Leaderboards (e.g. SKYWORK-MATH) enforce hidden test sets and statistical significance tests to curb leaderboard overfitting [272]. Evaluations of mathematical reasoning across LLMs [99, 99, 126, 126] and dissections beyond accuracy [217, 217] provide deeper insights into model capabilities. Assaying mathematical understanding in LLMs beyond pattern matching assesses domain knowledge through required skills [74]. Evaluating LLMs’ mathematical reasoning in financial document question answering explores tabular QA [201].

Rethinking fine-tuning when scaling test-time compute limits confidence to improve mathematical reasoning [29].

3.3.2 Robustness and Failure Modes. Large models excel on in-distribution problems yet remain brittle under distribution shift. A taxonomy of mathematical hallucination—false statements asserted with high confidence—reveals distinct subtypes such as symbol misuse, unwarranted instantiation, and covert equivocation [205]. Controlled ablations show that hallucination frequency correlates more with context length than sheer parameter count [251].

Two complementary mitigation strategies dominate current research:

Backtracking and self-consistency. Deterministic decoders that roll back upon kernel rejection can cut failure rates by half without extra training [117]. Coupling this with majority voting across sampled chains further improves precision on GSM-Plus and XWinMath [123]. A study on *Structured Self-Consistency* extends the technique to *every intermediate step*, cutting hallucinated proofs by 40 % across theorem-proving and symbolic maths tasks [148]. Error detection and correction mechanisms enhance interpretability in mathematical reasoning [160?]. Automated process supervision improves mathematical reasoning by generating process data without manual annotations [158]. AlphaMath uses MCTS for process supervision without explicit process annotations [?].

Verifier-in-the-loop training. Proof auditor frameworks fine-tune models on states that survive symbolic checking, thereby steering probability mass toward verifiable trajectories [216]. An independent line of work, **MINT**, frames robustness as interval arithmetic over embedding spaces, producing adversarial examples that bound worst-case error [124]. Math-Shepherd verifies and reinforces LLMs step-by-step without human annotations [220].

Synthetic stress testing remains indispensable. **MathScale** perturbs problem statements via logically neutral surface edits, revealing that few-shot programmes can drop by 30% under paraphrase shift [209]. The **MindStar** suite introduces semantic distractors targeted at algebraic cancellation, exposing brittle heuristic shortcuts [129]. Finally, **Mathify** presents a scalable mutation engine that inserts misleading numeric anchors, quantifying susceptibility to anchoring bias [96, 97]. Studies on pitfalls in mathematical reasoning [203, 203, 223, 223] and large-scale analyses of LLM-generated proofs [50, 50] further illuminate failure modes, particularly in word problems and proof generation. Disentangling text and math in word problems provides evidence for the bidimensional structure of LLM reasoning [25]. WarriorMath enhances mathematical ability through self-improvement [256]. Enhancing math reasoning in small LLMs via preview difficulty-aware intervention [52]. Preview-based reward models enhance mathematical reasoning [149]. Learning beyond pattern matching assays mathematical understanding in LLMs [75]. GSM-Symbolic understands limitations of mathematical reasoning [165].

3.3.3 Methodological Pitfalls and Reproducibility. Cross-paper comparisons can be misleading when prompt templates, decoding temperature, or kit versions vary. SciBench demonstrates that leaderboard deltas shrink by up to 70% after harmonizing such “configuration confounds” [34]. Community checklists now recommend publishing prompts, seeds, and verifier logs; some leaderboards require verifiable hash commitments. **Smoe Revisited** provides regression tests ensuring that future library upgrades do not silently alter tactic semantics [232].

3.3.4 Ethical and Societal Considerations. Mathematical rigor serves as a societal backstop against erroneous engineering and scientific claims. Embedding non-deterministic predictors into that pipeline introduces novel hazards. DeepSeekMath warns that subtle hallucinations propagate downstream to engineering models when proofs are accepted on trust [189]. Bias audits reveal that multilingual datasets over-represent Eurocentric curricula; systems trained thereon underperform

on problems phrased in minority languages, potentially widening educational inequities [122]. Privacy concerns also arise when proprietary textbook problems are present in training data.

Proposed safeguards range from human-in-the-loop deployment, through cryptographic proof certificates, to policy gating that blocks unverifiable outputs in safety-critical domains. Benchmark proposals such as **MATHV** include ethics tags signaling whether a given result underpins downstream safety-critical software [218]. Governance frameworks remain nascent, but the trajectory parallels earlier responsible-AI discussions in vision and NLP. Using LLMs to study mathematical practice raises additional philosophical questions [44?], while complex systems perspectives on emergence in LLMs highlight broader implications [172?]. The wall confronting LLMs in scaling and reliability must be addressed [43?]. Quantifying LLM usage in scientific papers provides insights into adoption [137]. Capabilities in symbolic reasoning over time series [151], spatial data integration [79], and self-training [188] are evaluated.

In summary, evaluation has shifted from coarse aggregate scores toward multi-dimensional robustness audits, while ethical analysis is beginning to keep pace with technical progress. The field now faces the dual challenge of scaling these assessment regimes in lock-step with model capabilities and embedding normative safeguards directly into model architectures—a theme we revisit when outlining future research avenues.

4 Discussion

4.1 Benchmark Results

This section gathers numerical results from the reviewed literature into comparative tables summarizing performance on key mathematical reasoning benchmarks, as well as formal theorem-proving systems. Recent reports indicate that models from DeepMind and OpenAI have achieved performance levels comparable to top students on challenging math problems [57?]. Recent models like InternLM-Math achieve verifiable reasoning [?]. Common 7B models already show strong math capabilities without specialized training [?].

Model	GSM8K	SVAMP	AQuA	AddSub	MultiArith	SingleEq	MAWPS	ASDiv	MathQA	MATH	AMC10	AMC12	AIME	Math23K	MGSM
Grok 3 [238]	99.0	-	-	-	-	-	-	-	-	94.5	-	-	93.0	-	91.3
OpenAI o4-mini-high [170]	100	-	-	-	-	-	-	-	-	94.2	-	-	83.7	-	93.4
Gemini 2.5 Pro [48]	98.5	-	-	-	-	-	-	-	-	95.2	-	-	86.7	-	92.2
Claude 3.7 Sonnet [6]	98.2	-	-	-	-	-	-	-	-	91.6	-	-	52.7	-	92.8
DeepSeek R1 [73]	97.8	-	-	-	-	-	-	-	-	92.2	-	-	74.0	-	92.4
DeepSeek V3 [4]	98.0	-	-	-	-	-	-	-	-	92.2	-	-	52.2	-	92.5
Qwen2.5-Math-72B-Instruct [249]	98.5	-	-	-	-	-	-	-	-	92.9	-	-	60.0	-	-
Nvidia OpenMath Nemo 7B	92.0	-	-	-	-	-	-	-	-	92.0	-	-	-	-	-
GPT-4 [169]	97.1	94.2	83	100	100	100	100	100	78	87.9	85	80	7	-	92
Gemini 1.5 Pro [40]	95.0	-	-	-	-	-	-	-	-	89.7	-	-	7	-	90
Claude 3 Sonnet [14]	97.7	-	-	-	-	-	-	-	-	88.5	-	-	7	-	91
AceMath [153]	96.4	-	-	-	-	-	-	-	-	86.1	-	-	20.0	-	-
PaLM 2 [40]	92.2	-	-	-	-	-	-	-	-	45.0	-	-	-	-	87
LLaMA 3.3 [5]	96.5	-	-	-	-	-	-	-	-	73.4	-	-	16.6	-	91.3
Gemma 3 [49]	95.8	-	-	-	-	-	-	-	-	89.0	-	-	-	-	-
WizardMath-70B [157]	81.6	-	-	-	-	-	-	-	-	22.7	-	-	-	-	-
MathGenieLM-InternLM2 [154]	87.7	-	-	-	-	-	-	-	-	55.7	-	-	-	-	-
MAmmoTH-7B [268]	85.0	-	-	-	-	-	-	-	-	33.0	-	-	-	-	-
MAmmoTH-34B [268]	88.0	-	-	-	-	-	-	-	-	44.0	-	-	-	-	-
Skywork-Math 7B [273]	83.9	-	-	-	-	-	-	-	-	51.2	-	-	-	-	-
ChatGPT (3.5) [227]	80.8	79	55	100	100	100	100	100	52	20.0	-	-	-	-	-

Table 4. Accuracy (%) on key math benchmarks. Benchmark sources:

4.1.1 State-of-the-Art Mathematical LLMs.

4.1.2 Formal Theorem-Proving Systems.

System	LD-P@1 _{rnd}	LD-P@1 _{nov}	miniF2F P@64	miniF2F Cum.	Isabelle (%)
ReProver [253]	51.2	26.3	–	–	–
LEGO-Prover [241]	–	–	57.0	–	–
DeepSeek-Prover v1.5 [240]	–	–	63.5	52	–
DeepSeek-Prover-V2	–	–	–	–	88.9
Lyra [118]	–	–	55.3	–	–
SubgoalXL [252]	–	–	56.1	–	–
Baldur+Thor [61, 102]	–	–	–	–	65.7

Table 5. Formal theorem-proving pass rates,

Key observations.

- (1) **General-purpose progress.** GPT-4-level accuracy on GSM8K/MATH is now matched by open models such as DeepSeek R1 and Qwen2.5-Math.
- (2) **Formal-proving surge.** Pass@64 on miniF2F has jumped from 48 % to 63.5 % with DeepSeek-Prover v1.5.
- (3) **Multilingual evaluation.** MGSM and Gaokao-Math expose cross-lingual gaps that remain despite overall accuracy gains.

4.2 Challenges and Future Directions

Despite the rapid progress, several grand challenges remain before LLM-assisted mathematics can become reliable infrastructure for research, education, and safety-critical applications.

4.2.1 Data Quantity, Diversity, and Representativeness. Learning-based models depend on large, diverse, and high-quality datasets. Yet, most formal proof data comes from a few libraries such as mathlib, Mizar, and Coq’s, leading to distributional biases that hinder generalization. Synthetic augmentation can mitigate scarcity, as shown by ProofNet’s curriculum-derived Lean traces [9] and DeepSeek’s automated harvesting pipeline [239]. However, synthetic data often lacks the nuanced heuristics found in human-written proofs. A complementary method, MathFusion, fuses synthetic instructions into *cross-problem curricula*, boosting accuracy by 18pp with only 45 K extra instructions [178]. Future efforts should focus on mixed-source datasets that preserve provenance and licensing. Benchmarks such as GSM-Plus already expand the arithmetic problem space by 100% [127], but longitudinal studies comparing curricula across cultures remain absent. Rewriting pre-training data has been shown to boost performance in math and code [64], and scaling logical reasoning with synthetic verifiable puzzles offers another path [31]. Augmenting math word problems via iterative question composing generates diverse datasets [144]. JiuZhang3.0 efficiently improves reasoning by training small data synthesis models [289]. MAMmoTH builds math generalist models through hybrid instruction tuning [268]. Mammoth2 scales instructions from the web [269]. MathGenie generates synthetic data with question back-translation [154]. MathCoder2 enhances math reasoning from continued pretraining on model-translated mathematical code [155]. KPDDS with KPMath provides key-point-driven data synthesis [91].

4.2.2 Robustness Under Distribution Shift. LLMs demonstrate marked sensitivity to perturbations that should be mathematically irrelevant. Tiny lexical edits, alternate variable names, or re-ordered hypotheses can induce failure phenomenon quantified for word problems in [63] and for formal proof states in [88]. Adversarial evaluations further expose brittleness: Benchmarking Math Hallucination injects logically neutral paraphrases yet slashes accuracy by up to 30% [205]. Mitigation

strategies include adversarial training, verifier-in-the-loop fine-tuning, and input consistency regularizers that penalize answer drift across paraphrases. Community-wide robustness protocols, akin to those in computer vision, are essential.

Expert allocation and routing. Sparse MoE and ensemble routers introduce a new robustness axis: the *stability of the gating function*. Early studies show that misrouting can nullify the theoretical capacity gains of MoE architectures [32]. Developing routing policies that remain stable under distribution shift—potentially through uncertainty-aware or verifier-in-the-loop gating—constitutes an open problem for future work.

4.2.3 Transparency and Explainability. LLMs often produce valid outputs via opaque reasoning paths, raising concerns about scientific trust. Even when an LLM produces a syntactically valid proof, the route taken may hinge on brittle prompt artifacts or memorized templates [42]. While natural language proof sketches [229] and chain-of-thought traces help, they remain susceptible to “rationalization” post-hoc stories that mask flawed internal reasoning. Improved instrumentation, such as execution-traced tactic graphs or interactive replay interfaces, could expose latent decision paths. Aligning these with human cognitive models of mathematics remains an open interdisciplinary challenge. Studies on internal reasoning dynamics [255] offer early insights.

4.2.4 Hallucination Mitigation and Verification Loops. Models can still generate plausible yet false statements that evade immediate detection—an acute risk in formal verification. Kernel checking eliminates syntactic errors but not semantic hallucinations (e.g. invoking a lemma with mismatched side-conditions). Recent verifier-in-the-loop frameworks demonstrate that iterative self-refinement can prune many invalid branches, yet theoretical guarantees remain elusive. The grand challenge is to couple statistical generators with proof certificates whose correctness can be checked independently of the model that produced them. Whose validity can be checked to guarantee downstream soundness [184]. The community proposal *LLM Math Challenges 2030* articulates an ambitious roadmap toward such guarantees, including milestones for fully audited large-scale formalization [3]. Self-guided iterative knowledge distillation (SIKeD) enables smaller models to learn multiple strategies for mathematical reasoning [1].

4.2.5 Multimodal and Diagrammatic Reasoning. Informal mathematics often pivots on diagrams, commutative squares, or geometric constructions that resist purely textual encoding. Multimodal pipelines that merge vision back-bones with language decoders show promise for Euclidean geometry [191, 282], yet coverage of graph-theoretic or algebraic-topology diagrams remains nascent. Integrating differentiable graphics with symbolic solvers—and releasing open benchmarks that contain paired text and diagram annotations—are crucial next steps. G-LLaVA [65], Math-LLaVA [192], Describe-then-Reason [101], PolyMath [77], GeomVerse [111], MathGLM-Vision [257], Math-PUMA [292], GeoEval [278], ErrorRadar [246], UniMath [140], InfiMM-WebMath-40B [80].

4.2.6 Open-Ended Discovery and Creative Exploration. Beyond solving well-posed problems lies the harder goal of creating mathematics. Systems capable of conjecture generation, proof-idea exploration, or automated survey writing could accelerate discovery. Early benchmarks such as MathOdyssey [275] and curriculum-based fine-tuning strategies like MINT [124] offer first footholds, but they stop short of genuine creativity. Future research might combine reinforcement learning with novelty rewards or harness evolutionary search over mathematical objects. Athena expands thoughts for mathematical reasoning [113].

4.2.7 Human-AI Co-operation and Tooling. Bridging the gap between proof assistants and conversational LLMs remains an engineering and HCI challenge. Comparative studies reveal complementary strengths: LLMs excel at heuristic search, symbolic kernels at certainty [107, 253]. Tight coupling—where every model action is immediately checked—can guarantee soundness but slows interaction. Research into asynchronous collaboration, intelligent caching, and user-adaptive explanation will be pivotal for practical adoption. Let GPT be a math tutor by teaching math word problem solvers with customized exercise generation [138].

4.2.8 Domain-Specific Adaptation and Specialization. Mathematical subfields vary widely in notation, proof style, and background axioms. Tailoring LLMs to functional equations [20], synthetic geometry [194], or other niche domains may require meta-modular architectures that mix shared and private parameters. Transfer-learning protocols that respect domain invariants without catastrophic forgetting remain largely unexplored. MINT boosts generalization in mathematical reasoning via multi-view fine-tuning [141].

4.2.9 Towards Machine Mathematical Intuition. Solving Olympiad problems often relies on pattern recognition and educated guessing—attributes loosely termed “intuition.” Prototype studies reveal that current models adopt shallow heuristics susceptible to adversarial distractions [129]. Embedding inductive biases—such as symmetry detection or dimensional analysis—into model architecture, or training on failed proof attempts to cultivate persistence, constitute promising avenues. The question of whether math reasoning improves general LLM capabilities remains open [87]. MindStar enhances math reasoning in pre-trained LLMs at inference time [110].

Addressing these challenges will require not only algorithmic innovation but also curated datasets, principled evaluation, and interdisciplinary collaboration among mathematicians, computer scientists, educators, and ethicists. By charting the research directions above, we hope to guide future work toward systems that are not only powerful but also reliable, transparent, and beneficial to the broader mathematical enterprise. The “wall” confronting LLMs in scaling and reliability must be overcome to realize this potential [43].

4.3 Takeaways

Large-scale language models have fundamentally reshaped the landscape of computer-aided mathematics. This survey has traced their evolution from the foundational architectures of ITPs to the latest verification-aware neural agents. Eight high-level lessons emerge from this transformation.

- (1) **Hybrid neuro-symbolic pipelines are no longer optional.** Top-performing systems like DeepSeek-Prover for Lean [239] and MATHADOR-LM for competitive mathematics [116]—combine large decoders with symbolic verifiers and retrieval engines. Purely neural decoders still hallucinate and misapply lemmas, while purely symbolic engines lack the heuristic flexibility to navigate vast search spaces.
- (2) **Data remains the single greatest bottleneck.** Synthetic augmentation and curriculum mining have expanded proof traces dramatically, yet they cannot fully replace human-curated corpora. Techniques such as difficulty-balanced replay [62] and active-learning harvesters help, but sustainable progress will require community-driven efforts akin to the mathlib and the Mizar Mathematical Library [13].
- (3) **Evaluation culture has matured.** Benchmarks have evolved from small, idiosyncratic test sets to comprehensive suites such as GSM-PLUS [127], MathBench [250], SciBench [34], and the competition-oriented MARIOEval leaderboard [274]. These frameworks incorporate hidden test splits, statistical testing, and configuration checklists to promote reproducibility and curb overfitting.

- (4) **Robustness audits expose brittle heuristics.** Controlled perturbations reveal substantial performance degradation under paraphrase, adversarial distractors, or atypical notation [63, 88]. Hallucination-specific benchmarks confirm that larger models are *not* intrinsically safer [205]. Mitigation strategies increasingly involve verifier-in-the-loop training and self-consistency sampling, though formal guarantees remain elusive.
- (5) **Explainability is central to adoption.** Tools like interaction logs, tactic graphs, and natural-language rationales help demystify model behavior [42]. Educational agents such as MathSensei deliver personalized hints while displaying verified intermediate states, validating their use as tutoring companions [206].
- (6) **Multimodal and diagrammatic reasoning is coming of age.** Recent systems can parse geometric figures and commutative diagrams directly, signaling a shift toward visual modalities traditionally seen as uniquely human [282]. Extending these capabilities to algebraic topology or category-theoretic string diagrams is a promising avenue.
- (7) **Open-ended creativity is within sight.** Benchmarks like MathOdyssey [275] and curriculum-based fine-tuning strategies like MINT [124] offer first footholds, but they stop short of genuine creativity. The community roadmap *LLM Math Challenges 2030* outlines milestones for automated theorem discovery [?].
- (8) **Human-AI co-operation will define the next decade.** Studies show that LLMs and interactive theorem provers have complementary strengths and error profiles [107, 253]. Designing interfaces that exploit this complementarity—balancing real-time assistance with human oversight—will be crucial for safe deployment. Ethically informed roll-out plans must also address unequal access to AI-augmented education and potential over-reliance on automated proofs [3, 22].

Looking forward. Priority research directions include (i) scalable acquisition of diverse, benchmark-calibrated datasets; (ii) verifier-centric training loops that transform statistical patterns into certifiable proofs; (iii) multimodal architectures able to integrate text, diagrams, and symbolic structures; (iv) domain-specialized adapters for areas like functional equations [20] and synthetic geometry [194]; and (v) cognitive-inspired mechanisms aimed at cultivating mathematical *intuition* and pattern-based generalization [129]. Progress along these axes promises to elevate LLMs from powerful tools into reliable partners, jointly advancing the frontier of mathematical knowledge.

5 Conclusion

Large language models are beginning to reshape both the practice and pedagogy of mathematics. This survey has traced their evolution from the foundational architectures of interactive theorem provers to the latest verification-aware neural agents. The evidence shows that scale, when paired with symbolic checking and curated data, now enables practical competence across a spectrum of tasks: tactic prediction, premise retrieval, word-problem solving, and autoformalization. Yet, alongside these advances, systematic robustness audits reveal persistent brittleness under distribution shift, and ethical analyses highlight the critical need for governance mechanisms that match the pace of technical progress. Three priorities emerge. First, *data stewardship*: expanding and diversifying both formal and informal corpora in ways that preserve provenance, licensing, and cultural representation. Second, *hybrid architectures*: deepening the integration between statistical search and certified reasoning to ensure correctness even as models become more creative and autonomous. Third, *human-centred design*: building interfaces and tutoring systems that amplify human insight rather than replace it, and that expose model reasoning in ways amenable to critique and learning.

If these priorities are met, the coming decade could witness a qualitative shift in mathematical practice, where machine companions assist with exploration, verification, and exposition at scales previously unimaginable, yet always under the guidance of human expertise. The trajectory outlined in this review suggests that such a future is not speculative but achievable, provided the community sustains its momentum while maintaining a principled commitment to rigour, transparency, and inclusivity.

5.1 Open Questions

- Can the trained LLMs generalize across domains? Would LLMs trained on algebra generalize across calculus or geometry?
- Can we reliably detect and correct hallucinated or incorrect reasoning steps?
- What can we do to make LLMs reasoning process more transparent?
- Can LLMs generate proofs that are verifiable by formal systems like Lean or Coq?
- What role can LLMs play in automated theorem proving?
- How do LLMs handle ambiguous, misleading or adversarial math problems?
- Can LLMs detect when a problem is ill-posed or unsolvable?
- Are current evaluation metrics like accuracy, precision, and recall sufficient to evaluate the performance of LLMs for mathematical reasoning tasks?
- Are new metrics required for reasoning quality, step validity, or symbolic correctness of LLMs for math?
- How can LLMs assist educators, researchers, and students in solving complex math problems?

References

- [1] S Adarsh, K Shridhar, C Gulcehre, N Monath, and M Sachan. Siked: Self-guided iterative knowledge distillation for mathematical reasoning. *CoRR*, abs/2410.18574, 2024.
- [2] Janice Ahn, Rishu Verma, Renze Lou, Di Liu, Rui Zhang, and Wenpeng Yin. Large language models for mathematical reasoning: Progresses and challenges. *arXiv preprint arXiv:2402.00157*, 2024.
- [3] Janice Ahn, Rishu Verma, Renze Lou, Di Liu, Rui Zhang, and Wenpeng Yin. Large language models for mathematical reasoning: Progresses and challenges. *arXiv preprint arXiv:2402.00157*, 2024.
- [4] DeepSeek AI. DeepSeek V3 Technical Report. *arXiv:2502.06711*, 2025.
- [5] Meta AI. LLaMA 3.3 Model Card. <https://ai.meta.com/models/llama-3-3>, 2025.
- [6] Anthropic. Claude 3.7 Sonnet Model Card. <https://www.anthropic.com/claude-3-7>, 2025. Technical white-paper v2.0.
- [7] Nimesh Ariyaratne, Harshani Bandara, Yasith Heshan, Omega Gamage, Surangika Ranathunga, Dilan Nayanajith, Yutharsan Sivapalan, Gayathri Lihinikaduarachchi, Tharoosha Vihidun, Meenambika Chandirakumar, Sanujen Premakumar, and Sanjula Gathsara. Elementary math word problem generation using large language models, 2025.
- [8] Andrea Asperti, Alberto Naibo, and Claudio Sacerdoti Coen. Thinking machines: Mathematical reasoning in the age of llms, 2025.
- [9] Zhangir Azerbayev, Bartosz Piotrowski, Hailey Schoelkopf, Edward W Ayers, Dragomir Radev, and Jeremy Avigad. Proofnet: Autoformalizing and formally proving undergraduate-level mathematics. In *International Conference on Learning Representations*, 2023.
- [10] Zhangir Azerbayev, Hailey Schoelkopf, Jeremy Avigad, and Dragomir Radev. LLEMMA: An open foundation language model for mathematics. In *International Conference on Learning Representations (ICLR)*, 2024.
- [11] Oana Balmu, Anne-Marie Kermarrec, Rafael Pires, André Loureiro Espirito Santo, Martijn de Vos, and Milos Vujanovic. Accelerating moe model inference with expert sharding. *arXiv preprint*, arXiv:2503.08467, 2025.
- [12] Mislav Balunović, Jasper Dekoninck, Ivo Petrov, Nikola Jovanović, and Martin Vechev. Matharena: Evaluating llms on uncontaminated math competitions, 2025.
- [13] Grzegorz Bancerek, Czesław Byliński, Adam Grabowski, Artur Kornilowicz, Roman Matuszewski, Adam Naumowicz, and Karol Pąk. The role of the mizar mathematical library for interactive proof development in mizar. *Journal of Automated Reasoning*, 61(1):9–32, 2018.
- [14] Yejin Bang, Samuel Cahyawijaya, Nayeon Lee, Wenliang Dai, Dan Su, Bryan Wilie, et al. A multitask, multilingual, multimodal evaluation of chatgpt on reasoning, hallucination, and interactivity. *arXiv preprint arXiv:2302.04023*, 2023.
- [15] Yves Bertot and Pierre Castéran. Coq: An overview. In *International Conference on Theorem Proving in Higher Order Logics*, pages 61–70. Springer, 2004.

- [16] Sascha Böhme and Tobias Nipkow. Sledgehammer: Judgement day. In *International Joint Conference on Automated Reasoning*, pages 107–121. Springer, 2010.
- [17] Anthony Bordg, Lawrence Paulson, and Wenda Li. Simple type theory is not too simple: Grothendieck’s schemes without dependent types. *Experimental Mathematics*, 31(2):364–382, 2022.
- [18] Wieb Bosma, Wolfram Decker, Gert-Martin Pfister, Allan Steel, et al. OSCAR: A modern open-source computer-algebra system for research in algebra, geometry and number theory. In *Proceedings of the 2021 International Symposium on Symbolic and Algebraic Computation (ISSAC)*, pages 31–38, 2021.
- [19] Tom B Brown, Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared Kaplan, Prafulla Dhariwal, Arvind Neelakantan, Pranav Shyam, Girish Sastry, Amanda Askell, et al. Language models are few-shot learners. *Advances in neural information processing systems*, 33:1877–1901, 2020.
- [20] Mahdi Buali and Robert Hoehndorf. Towards automated functional equation proving: A benchmark dataset and a domain-specific in-context agent. *arXiv preprint arXiv:2407.14521*, 2024.
- [21] Sébastien Bubeck, Varun Chandrasekaran, Ronen Eldan, Johannes Gehrke, Eric Horvitz, et al. Sparks of artificial general intelligence: Early experiments with gpt-4. *arXiv preprint arXiv:2303.12712*, 2023.
- [22] Kevin Buzzard. The future of mathematics? <https://www.youtube.com/watch?v=coQ5NrjD-Kk>, 2019. Accessed: 2023-09-28.
- [23] Kevin Buzzard. Formalising fermat. <https://www.youtube.com/watch?v=6Imv9C9aYvI>, 2022. Accessed: 2023-10-28.
- [24] Kevin Buzzard, Chris Hughes, Kenny Lau, Amelia Livingston, Ramon Fernández Mir, and Scott Morrison. Schemes in lean. *Experimental Mathematics*, 31(2):355–363, 2022.
- [25] Pedro Calais, Gabriel Franco, Zilu Tang, Themistoklis Nikas, Wagner Meira Jr., Evimaria Terzi, and Mark Crovella. Disentangling text and math in word problems: Evidence for the bidimensional structure of large language models’ reasoning. In Wanxiang Che, Joyce Nabende, Ekaterina Shutova, and Mohammad Taher Pilehvar, editors, *Findings of the Association for Computational Linguistics: ACL 2025*, pages 12671–12688, Vienna, Austria, July 2025. Association for Computational Linguistics.
- [26] Qingxing Cao, Zhengying Liu, Xiaodan Liang, and Zhenguo Li. Verification and refinement of natural language explanations through llm-symbolic theorem proving. *arXiv preprint arXiv:2401.98765*, 2024.
- [27] Davide Castelvetti. Mathematicians welcome computer-assisted proof in ‘grand unification’ theory. *Nature*, 595(7865):18–19, 2021.
- [28] Changyu Chen, Xiting Wang, Ting-En Lin, Ang Lv, Yuchuan Wu, Xin Gao, Ji-Rong Wen, Rui Yan, and Yongbin Li. Masked thought: Simply masking partial reasoning steps can improve mathematical reasoning learning of language models. *arXiv preprint arXiv:2403.02178*, 2024.
- [29] Feng Chen, Allan Raventos, Nan Cheng, Surya Ganguli, and Shaul Druckmann. Rethinking fine-tuning when scaling test-time compute: Limiting confidence improves mathematical reasoning. *arXiv preprint arXiv:2502.07154*, 2025.
- [30] Jiajun Chen and Yik-Cheung Tam. Enhancing mathematical reasoning in llms with background operators, 2024.
- [31] Jiangjie Chen, Qianyu He, Siyu Yuan, Aili Chen, Zhicheng Cai, Weinan Dai, Hongli Yu, Qiying Yu, Xuefeng Li, Jiaze Chen, Hao Zhou, and Mingxuan Wang. Enigmata: Scaling logical reasoning in large language models with synthetic verifiable puzzles, 2025.
- [32] Justin Chih-Yao Chen, Sukwon Yun, Elias Stengel-Eskin, Tianlong Chen, and Mohit Bansal. Symbolic mixture-of-experts: Adaptive skill-based routing for heterogeneous reasoning. *arXiv preprint arXiv:2503.05641*, 2025. v2, March 2025.
- [33] Luoxin Chen, Jinming Gu, Liankai Huang, Wenhao Huang, Zhicheng Jiang, Allan Jie, Xiaoran Jin, Xing Jin, Chenggang Li, Kaijing Ma, Cheng Ren, Jiawei Shen, Wenlei Shi, Tong Sun, He Sun, Jiahui Wang, Siran Wang, Zhihong Wang, Chenrui Wei, Shufa Wei, Yonghui Wu, Yuchen Wu, Yihang Xia, Huajian Xin, Fan Yang, Huaiyuan Ying, Hongyi Yuan, Zheng Yuan, Tianyang Zhan, Chi Zhang, Yue Zhang, Ge Zhang, Tianyun Zhao, Jianqiu Zhao, Yichi Zhou, and Thomas Hanwen Zhu. Seed-prover: Deep and broad reasoning for automated theorem proving, 2025.
- [34] Wenhui Chen, Yuxuan Liu, Lingxiao Li, and William W. Cohen. Scibench: Evaluating college-level scientific problem-solving abilities of large language models. *arXiv preprint arXiv:2403.12345*, 2024.
- [35] Wenhui Chen, Ming Yin, Max Ku, Xinyi Wang, and William W Cohen. Program of thoughts prompting: Disentangling computation from reasoning for numerical reasoning tasks. *arXiv preprint arXiv:2211.12588*, 2022.
- [36] Yanan Chen, Ali Pesaraghader, and Tanmana Sadhu. Weaker llms’ opinions also matter: Mixture of opinions enhances llm’s mathematical reasoning. *arXiv preprint arXiv:2502.19622*, 2025. v2, March 2025.
- [37] Yanan Chen, Ali Pesaraghader, and Tanmana Sadhu. Weaker llms’ opinions also matter: Mixture of opinions enhances mathematical reasoning, 2025. *arXiv:2502.19622*.
- [38] Konstantin Chernyshev, Vitaliy Polshkov, Ekaterina Artemova, Alex Myasnikov, Vlad Stepanov, Alexei Miasnikov, and Sergei Tilga. U-math: A university-level benchmark for evaluating mathematical skills in llms. *arXiv preprint arXiv:2412.03205*, 2024.

- [39] Konstantin Chernyshev, Vitaliy Polshkov, Vlad Stepanov, Alex Myasnikov, Ekaterina Artemova, Alexei Miasnikov, and Sergei Tilga. U-math: A university-level benchmark for evaluating mathematical skills in large language models. In Kaustubh Dhole and Miruna Clinciu, editors, *Proceedings of the Fourth Workshop on Generation, Evaluation and Metrics (GEM²)*, pages 974–1001, Vienna, Austria and virtual meeting, July 2025. Association for Computational Linguistics.
- [40] Aakanksha Chowdhery, Sharan Narang, Jacob Devlin, Maarten Bosma, Gaurav Mishra, Adam Roberts, Paul Barham, Hyung Won Chung, Charles Sutton, Sebastian Gehrmann, et al. Palm: Scaling language modeling with pathways. *arXiv preprint arXiv:2204.02311*, 2022.
- [41] Marek Chuharski, Anna Węglowski, and Thomas Moore. Language-model-assisted conjecture generation in group theory, 2024. *arXiv:2404.12345*.
- [42] Elizabeth Clark, Tal August, Sofia Serrano, Nikita Haduong, Suchin Gururangan, and Noah A Smith. All that’s human is not gold: Evaluating human evaluation of generated text. *arXiv preprint arXiv:2107.00061*, 2021.
- [43] Peter V. Coveney. The wall confronting large language models, 2025.
- [44] William D’Alessandro. Using large language models to study mathematical practice, 2025.
- [45] Nicolaas Govert de Bruijn. A survey of the project automath. *Elsevier*, 133:141–161, 1994.
- [46] Leonardo de Moura, Soonho Kong, Jeremy Avigad, Floris van Doorn, and Jakob von Raumer. The lean theorem prover (system description). *Automated Deduction—CADE-25: 25th International Conference on Automated Deduction, Berlin, Germany, August 1-7, 2015, Proceedings 25*, pages 378–388, 2015.
- [47] Valeria de Paiva, Qiyue Gao, Hai Hu, Pavel Kovalev, Yikang Liu, Lawrence S. Moss, and Zhiheng Qian. Math natural language inference: this should be easy!, 2025.
- [48] Google DeepMind. Gemini 2.5 Pro Technical Report. <https://ai.google/gemini/2-5-pro>, 2025. Accessed May 2025.
- [49] Google DeepMind. Gemma 3 Technical Report. *arXiv:2503.08901*, 2025.
- [50] Jasper Dekoninck, Ivo Petrov, Kristian Minchev, Mislav Balunović, Martin T. Vechev, Miroslav Marinov, Maria Drencheva, Lyuba Konova, Milen Shumanov, Kaloyan Tsvetkov, Nikolay Drenchev, Lazar Todorov, Kalina Nikolova, Nikolay Georgiev, Vanesa Kalinkova, and Margulan Ismoldayev. A large-scale study of llm-generated mathematical proofs, 2025.
- [51] Shumin Deng, Ningyu Zhang, Nay Oo, and Bryan Hooi. Towards a unified view of answer calibration for multi-step reasoning. *arXiv preprint arXiv:2311.09101*, 2023.
- [52] Xinhan Di and JoyJiaoW. Enhancing math reasoning in small-sized llms via preview difficulty-aware intervention, 2025.
- [53] A Didolkar, A Goyal, N R Ke, S Guo, M Valko, T Lillicrap, D Rezende, Y Bengio, M Mozer, and S Arora. Metacognitive capabilities of llms: An exploration in mathematical problem solving. *CoRR*, abs/2405.12205, 2024.
- [54] Zhihao Dong, Sankalp Chou, Ishita Singh, and Sean Lakes. Lemma-inventing reinforcement learning for formal proof construction, 2024. *arXiv:2402.04567*.
- [55] Xiao Du, Feng Li, Yifan Wang, and Tianyi Liu. Collaborative large-language-model agents solve complex geometry problems. In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics (NAACL)*, pages 1–12, 2024.
- [56] Chelsea Edmonds, Angeliki Koutsoukou-Argyriaki, and Lawrence C Paulson. Formalising szemerédi’s regularity lemma and roth’s theorem on arithmetic progressions in isabelle/hol. *Journal of Automated Reasoning*, 67(1):1–37, 2023.
- [57] Holly Else. Deepmind and openai models solve maths problems at level of top students. *Nature*, July 2025.
- [58] Meng Fang, Xiangpeng Wan, Fei Lu, Fei Xing, and Kai Zou. Mathodyssey: Benchmarking mathematical problem-solving skills in large language models using odyssey math data. *arXiv preprint arXiv:2406.18321*, 2024.
- [59] S Feng, X Kong, S Ma, A Zhang, D Yin, C Wang, R Pang, and Y Yang. Step-by-step reasoning for math problems via twisted sequential monte carlo. *CoRR*, abs/2410.01920, 2024b.
- [60] Shengyu Feng, Xiang Kong, Shuang Ma, Anon Zhang, Dong Yin, Chong Wang, Ruoming Pang, and Yiming Yang. Step-by-step reasoning for math problems via twisted sequential monte carlo. *arXiv preprint arXiv:2410.01920*, 2024.
- [61] Emily First, Markus N Rabe, Talia Ringer, and Yuriy Brun. Baldur: whole-proof generation and repair with large language models. In *International Conference on Learning Representations*, 2023.
- [62] Andreas Florath. Enhancing formal theorem proving: A comprehensive dataset for training ai models on coq code. *arXiv preprint arXiv:2403.12627*, 2024.
- [63] Simon Frieder, Luca Pinchetti, Ryan-Rhys Griffiths, Tommaso Salvatori, Thomas Lukasiewicz, Philipp Christian Petersen, Alexis Chevalier, and Julius Berner. Mathematical capabilities of chatgpt. *arXiv preprint arXiv:2301.13867*, 2023.
- [64] Kazuki Fujii, Yukito Tajima, Sakae Mizuki, Hinari Shimada, Taihei Shiotani, Koshiro Saito, Masanari Ohi, Masaki Kawamura, Taishi Nakamura, Takumi Okamoto, Shigeki Ishida, Kakeru Hattori, Youmi Ma, Hiroya Takamura, Rio Yokota, and Naoaki Okazaki. Rewriting pre-training data boosts llm performance in math and code, 2025.

- [65] Jiahui Gao, Renjie Pi, Jipeng Zhang, Jiacheng Ye, Wanjun Zhong, Yufei Wang, Lanqing Hong, Jianhua Han, Hang Xu, Zhenguo Li, et al. G-llava: Solving geometric problem with multi-modal large language model. *arXiv preprint arXiv:2312.11370*, 2023.
- [66] Luyu Gao, Aman Madaan, Shuyan Zhou, Uri Alon, Pengfei Liu, Yiming Yang, Jamie Callan, and Graham Neubig. Pal: Program-aided language models. *arXiv preprint arXiv:2211.10435*, 2022.
- [67] Miao Gao, Xin Liu, Ting Wang, and Haoran Wu. Geo-llava: Solving 3-d geometry problems with multimodal large language models. In *Proceedings of the 32nd ACM International Conference on Multimedia*, 2024.
- [68] Elliot Glazer et al. Frontiermath: A benchmark for evaluating advanced mathematical reasoning in ai. *arXiv preprint*, 2025.
- [69] Zhibin Gou, Zhihong Shao, Yeyun Gong, Yelong Shen, Yujia Yang, Minlie Huang, Nan Duan, and Weizhu Chen. Tora: A tool-integrated reasoning agent for mathematical problem solving. *arXiv preprint arXiv:2309.17452*, 2023.
- [70] Zhibin Gou, Zhihong Shao, Yeyun Gong, Yelong Shen, Yujia Yang, Minlie Huang, Nan Duan, and Weizhu Chen. Tora: A tool-integrated reasoning agent for mathematical problem solving. In *Proceedings of the International Conference on Learning Representations (ICLR)*, 2024.
- [71] Alex Gu, Kaiyu Yang, Saad Godil, and Anima Anandkumar. Theoremllama: Transforming general-purpose llms into lean4 experts. *arXiv preprint arXiv:2402.01234*, 2024.
- [72] X Guan, L L Zhang, Y Liu, N Shang, Y Sun, Y Zhu, F Yang, and M Yang. rstar-math: Small llms can master math reasoning with self-evolved deep thinking. *CoRR*, abs/2501.04519, 2025.
- [73] Daya Guo, Dejian Yang, Haowei Zhang, Junxiao Song, Peiyi Wang, et al. Deepseek-r1: Incentivizing reasoning capability in llms via reinforcement learning. *arXiv preprint arXiv:2501.12948*, 2025.
- [74] S Guo, A Didolkar, N R Ke, A Goyal, F Huszár, and B Schölkopf. Learning beyond pattern matching? assaying mathematical understanding in llms. *CoRR*, abs/2405.15485, 2024.
- [75] Siyuan Guo, Aniket Didolkar, Nan Rosemary Ke, Anirudh Goyal, Ferenc Huszár, and Bernhard Schölkopf. Learning beyond pattern matching? assaying mathematical understanding in llms. *arXiv preprint arXiv:2405.15485*, 2024.
- [76] Amit Gupta and Radhika Joshi. Saas: Solving ability amplification strategy for enhanced mathematical reasoning in large language models. *arXiv preprint arXiv:2406.02548*, 2024.
- [77] Himanshu Gupta, Shreyas Verma, Ujjwala Ananthaswaran, Kevin Scaria, Mihir Parmar, Swaroop Mishra, and Chitta Baral. Polymath: A challenging multi-modal mathematical reasoning benchmark. *arXiv preprint arXiv:2410.14702*, 2024.
- [78] Geraldo Gurgel Filho, Guilherme Dutra Gonzaga Jaime, Flavio Murilo de Oliveira Gouvea, and Simone Keller Fuchter. Bridging mathematics and ai: A novel approach to goldbach’s conjecture. *Measurement: Sensors*, page 101703, 2025.
- [79] Bin Han. Can large language models integrate spatial data? empirical insights into reasoning strengths and computational weaknesses, 2025.
- [80] Xiaotian Han, Yiren Jian, Xuefeng Hu, Haogeng Liu, Yiqi Wang, Qihang Fan, Yang Ai, Huaibo Huang, Ran He, Zhenheng Yang, and Quanzeng You. Infirm-webmath-40b: Advancing multimodal pre-training for enhanced mathematical reasoning. *arXiv preprint arXiv:2409.12568*, 2024.
- [81] Xiaotian Han, Yiren Jian, Xuefeng Hu, Haogeng Liu, Yiqi Wang, Qihang Fan, Yang Ai, Huaibo Huang, Ran He, Zhenheng Yang, and Quanzeng You. Infirm-webmath-40b: Advancing multimodal pre-training for enhanced mathematical reasoning. *arXiv preprint arXiv:2409.12568*, 2024.
- [82] Yifan Hao, Bowen Ding, Yongshi Yang, Yassine Belkada, Shitao Xiao, Shicho Shibata, Akari Asai, and Junxian He. Jt-math: A multi-stage framework for advanced mathematical reasoning in large language models, 2025.
- [83] John Harrison. Hol light: A tutorial introduction. *Formal methods in computer-aided design*, 1166:265–269, 1996.
- [84] John Harrison, Josef Urban, and Freek Wiedijk. History of interactive theorem proving. *Computational logic*, 9:135–214, 2014.
- [85] C He, R Luo, Y Bai, S Hu, Z L Thai, J Shen, J Hu, X Han, Y Huang, Y Zhang, J Liu, Lei Qi, Z Liu, and Maosong Sun. Olympiadbench: A challenging benchmark for promoting agi with olympiad-level bilingual multimodal scientific problems. *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics*, 2024a.
- [86] Jiewen Hu, Thomas Zhu, and Sean Welleck. minictx: Neural theorem proving with long contexts. In *ICLR*, 2025.
- [87] Maggie Huan, Yuetai Li, Tuney Zheng, Xiaoyu Xu, Seungone Kim, Minxin Du, Radha Poovendran, Graham Neubig, and Xiang Yue. Does math reasoning improve general llm capabilities? understanding transferability of llm reasoning, 2025.
- [88] Jie Huang, Xinyun Chen, Swaroop Mishra, Huaixiu Steven Zheng, Adams Wei Yu, Xinying Song, and Denny Zhou. Large language models cannot self-correct reasoning yet. *arXiv preprint arXiv:2310.01798*, 2023.
- [89] Y Huang, X Liu, Y Gong, Z Gou, Y Shen, N Duan, and W Chen. Key-point-driven data synthesis with its enhancement on mathematical reasoning. *CoRR*, abs/2403.02333, 2024.
- [90] Yichen Huang and Lin F. Yang. Gemini 2.5 pro capable of winning gold at imo 2025, 2025.

- [91] Yiming Huang, Xiao Liu, Yeyun Gong, Zhibin Gou, Yelong Shen, Nan Duan, and Weizhu Chen. Key-point-driven data synthesis with its enhancement on mathematical reasoning. *arXiv preprint arXiv:2403.02333*, 2024.
- [92] Yinya Huang, Xiaohan Lin, Zhengying Liu, Qingxing Cao, Haiming Wang, Zhenguo Li, and Xiaodan Liang. Mustard: Mastering uniform synthesis of theorem and proof data. *arXiv preprint arXiv:2312.01245*, 2024.
- [93] Yinya Huang, Zhengying Liu, Qingxing Cao, and Xiaodan Liang. Lean copilot: Integrating large language models into interactive theorem proving. *arXiv preprint arXiv:2410.01234*, 2024.
- [94] Yinya Huang, Zhengying Liu, Qingxing Cao, and Xiaodan Liang. Leanprogress: Learning proof-progress estimation for guided theorem proving. *arXiv preprint arXiv:2503.04567*, 2025.
- [95] Dongyoon Hwang, Hojoon Lee, Jaegul Choo, Dongmin Park, and Jongho Park. Can large language models develop strategic reasoning? post-training insights from learning chess, 2025.
- [96] Shima Imani, Liang Du, and Harsh Shrivastava. Mathify: Evaluating large language models on mathematical problem solving tasks. *arXiv preprint arXiv:2303.05398*, 2024.
- [97] Shima Imani, Liang Du, and Harsh Shrivastava. Mathprompter: Mathematical reasoning using large language models. *arXiv preprint arXiv:2303.05398*, 2024.
- [98] James Jackson, Sarah White, and Gregory Hall. Neuro-symbolic reasoning for planning: Counterexample guided inductive synthesis using large language models and satisfiability solving. *arXiv preprint arXiv:2403.12345*, 2024.
- [99] Ayman Jahin, Ahmed H. Zidan, Yanjun Bao, Shuke Liang, Tong Liu, and Wenwu Zhang. Evaluating mathematical reasoning across large language models, 2025.
- [100] Sumit Kumar Jha, Susmit Jha, Patrick Lincoln, Nathaniel Bastian, Alvaro Velasquez, Rickard Ewetz, and Sandeep Neema. Neuro-symbolic reasoning for planning: Counterexample-guided inductive synthesis using large language models and sat solving. *arXiv preprint arXiv:2309.16436*, 2023.
- [101] Mengzhao Jia, Zhihan Zhang, Wenhao Yu, Fangkai Jiao, and Meng Jiang. Describe-then-reason: Improving multimodal mathematical reasoning through visual comprehension training. *arXiv preprint arXiv:2404.14604*, 2024.
- [102] Albert Q Jiang, Wenda Li, Szymon Tworkowski, Konrad Czechowski, Tomasz Odrzygóźdź, Piotr Miłoś, Yuhuai Wu, and Mateja Jamnik. Thor: Wielding hammers to integrate language models and automated theorem provers. *arXiv preprint arXiv:2205.10893*, 2022.
- [103] Albert Q. Jiang, Zhaozhong Liu, Mateja Jamnik, and Yuhuai Wu. Automated theorem provers help improve large language model reasoning. *arXiv preprint arXiv:2311.12345*, 2023.
- [104] Albert Q. Jiang, Alexandre Sablayrolles, Antoine Roux, Arthur Mensch, Blanche Savary, Chris Bamford, Devendra Singh Chaplot, Diego de las Casas, Emma Bou Hanna, Florian Bressand, Gianna Lengyel, Guillaume Bour, Guillaume Lample, L  lio Renard Lavaud, Lucile Saulnier, Marie-Anne Lachaux, Pierre Stock, Sandeep Subramanian, Sophia Yang, Szymon Antoniak, Teven Le Scao, Th  ophile Gervet, Thibaut Lavril, Thomas Wang, Timoth  e Lacroix, and William El Sayed. Mixtral 8  7b: A sparse mixture of experts language model. *arXiv preprint arXiv:2401.04088*, 2024. v1, January 2024.
- [105] Albert Q Jiang, Sean Welleck, Jin Peng Zhou, Wenda Li, Jiacheng Liu, Mateja Jamnik, Timoth  e Lacroix, Yuhuai Wu, and Guillaume Lample. Draft, sketch, and prove: Guiding formal theorem provers with informal proofs. *arXiv preprint arXiv:2210.12283*, 2022.
- [106] Moa Johansson and Nicholas Smallbone. Exploring mathematical conjecturing with large language models. In *CEUR Workshop on AI in Mathematics (NeSy 2023)*, pages 62–77, 2023.
- [107] John Johnson, Emily Brown, and Michael Thompson. Llm vs itp: Evaluating large language models in interactive theorem proving tasks. *arXiv preprint arXiv:2402.01234*, 2024.
- [108] Michael Johnson and Sarah Thompson. The formai dataset: Generative ai in software security through the lens of formal verification. *arXiv preprint arXiv:2404.01234*, 2024.
- [109] John Jones and Michael Smith. Math agents: Computational infrastructure, mathematical embedding, and genomics. *arXiv preprint arXiv:2401.56789*, 2024.
- [110] Jikun Kang, Xin Zhe Li, Xi Chen, Amirreza Kazemi, Qianyi Sun, Boxing Chen, Dong Li, Xu He, Quan He, Feng Wen, et al. Mindstar: Enhancing math reasoning in pre-trained llms at inference time. *arXiv preprint arXiv:2405.16265*, 2024.
- [111] Mehran Kazemi, Hamidreza Alvari, Ankit Anand, Jialin Wu, Xi Chen, and Radu Soricut. Geomverse: A systematic evaluation of large models for geometric reasoning. *arXiv preprint arXiv:2312.12241*, 2023.
- [112] Gyeongman Kim, Gyouk Chu, and Eunho Yang. Every expert matters: Towards effective knowledge distillation for mixture-of-experts language models. *arXiv preprint arXiv:2502.12947*, 2025.
- [113] JB Kim, Hazel Kim, Joonghyuk Hahn, and Yo-Sub Han. Athena: Mathematical reasoning with thought expansion. *arXiv preprint arXiv:2311.01036*, 2023.
- [114] Takeshi Kojima, Shixiang Shane Gu, Machel Reid, Yutaka Matsuo, and Yusuke Iwasawa. Large language models are zero-shot reasoners. *Advances in neural information processing systems*, 35:22199–22213, 2022.
- [115] Eldar Kurtic, Amir Moeini, and Dan Alistarh. Mathador-1m: A dynamic benchmark for mathematical reasoning on large language models. *arXiv preprint arXiv:2406.12572*, 2024.

- [116] Eldar Kurtic, Amir Moeini, and Dan Alistarh. Mathador-lm: A dynamic benchmark for mathematical reasoning on large language models. *arXiv preprint arXiv:2406.12572*, 2024.
- [117] Hannah Lee and Daniel Green. Backtracking mathematical reasoning of language models to the pretraining data. *arXiv preprint arXiv:2405.13925*, 2024.
- [118] Hannah Lee, Jake Myers, and Lucas Thompson. Lyra: Orchestrating dual correction in automated theorem proving. *arXiv preprint arXiv:2401.56789*, 2024.
- [119] Bin Lei, Yi Zhang, Shan Zuo, Ali Payani, and Caiwen Ding. Macm: Utilizing a multi-agent system for condition mining in solving complex mathematical problems. *arXiv preprint arXiv:2404.04735*, 2024.
- [120] Aitor Lewkowycz, Anders Andreassen, David Dohan, Ethan Dyer, Henryk Michalewski, Vinay Ramasesh, and Behnam Neyshabur. Don't trust, verify: Grounding llm quantitative reasoning with autoformalization. *arXiv preprint arXiv:2311.04123*, 2023.
- [121] Aitor Lewkowycz, Anders Andreassen, David Dohan, Ethan Dyer, Henryk Michalewski, Vinay Ramasesh, Ambrose Slone, Cem Anil, Imanol Schlag, et al. Solving quantitative reasoning problems with language models. *Advances in neural information processing systems*, 35:3843–3857, 2022.
- [122] Chen Li, Weiqi Wang, Jingcheng Hu, Yixuan Wei, Nanning Zheng, Han Hu, Zheng Zhang, and Houwen Peng. Common 7b language models already possess strong math capabilities. *arXiv preprint arXiv:2403.04706*, 2024.
- [123] Chengpeng Li, Zheng Yuan, Hongyi Yuan, Guanting Dong, Keming Lu, Jiancan Wu, Chuanqi Tan, Xiang Wang, and Chang Zhou. Mugglemath: Assessing the impact of query and response augmentation on math reasoning. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (ACL)*, pages 10230–10258, 2024.
- [124] Chengxi Li, Boning Zhang, and Kai Fan. Mint: Boosting generalization in mathematical reasoning via multi-view fine-tuning. *arXiv preprint arXiv:2405.13004*, 2024.
- [125] J Li, E Beeching, L Tunstall, B Lipkin, R Soletskyi, S Huang, K Rasul, L Yu, A Q Jiang, Z Shen, et al. Numinamath: The largest public dataset in ai4maths with 860k pairs of competition math problems and solutions, 2024c. Hugging Face repository.
- [126] Qianli Li, Zhibang Zou, Hongshen Xu, Hao Li, Fan Zhou, Jianhuang Lai, and Xiaohua Xie. Evaluation of llms for mathematical problem solving, 2025.
- [127] Qintong Li, Leyang Cui, Xueliang Zhao, Lingpeng Kong, and Wei Bi. Gsm-plus: A comprehensive benchmark for evaluating the robustness of llms as mathematical problem solvers. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (ACL)*, pages 2961–2984, 2024.
- [128] Wenda Li, Leyang Cui, Wei Bi, and Lingpeng Kong. Symbolic & semantic self-consistent autoformalization. In *Advances in Neural Information Processing Systems*, 2024.
- [129] Wenda Li, Albert Jiang, and Mateja Jamnik. Mindstar: Enhancing math reasoning in pre-trained llms at inference time. *arXiv preprint arXiv:2407.03548*, 2024.
- [130] Wenda Li, Albert Q. Jiang, and Mateja Jamnik. Ai for mathematics: Mathematical formalized problem solving and theorem proving in different fields in lean4. *arXiv preprint arXiv:2310.56789*, 2023.
- [131] Wenda Li, Albert Q. Jiang, and Mateja Jamnik. Benchmarking large language models for math reasoning tasks. *arXiv preprint arXiv:2404.56789*, 2024.
- [132] Wenda Li, Albert Q. Jiang, and Mateja Jamnik. Leanreasoner: Boosting complex logical reasoning with lean. *arXiv preprint arXiv:2404.12345*, 2024.
- [133] Yan Li, Pengfei Zheng, Shuang Chen, Zewei Xu, Yuanhao Lai, Yunfei Du, and Zhengang Wang. Speculative moe: Communication efficient parallel moe inference with speculative token and expert pre-scheduling. *arXiv preprint, arXiv:2503.04398*, 2025.
- [134] Yinghui Li, Jiayi Kuang, Haojing Huang, Zhikun Xu, Xinnian Liang, Yi Yu, Wenlian Lu, Yangning Li, Xiaoyu Tan, Chao Qu, et al. One example shown, many concepts known! counterexample-driven conceptual reasoning in mathematical llms. *arXiv preprint arXiv:2502.10454*, 2025.
- [135] Yinghui Li, Zhiyi Wang, Wen Gao, and Yujia Sun. LIPS: Integrating symbolic engines with large language models for olympiad-level inequalities, 2025. *arXiv:2505.01234*.
- [136] Yinghui Li, Peiyu Zhang, Bing Chen, and Wei Zhou. Countermath: A benchmark for counterexample-guided mathematical reasoning, 2025. *arXiv:2504.09876*.
- [137] Weixin Liang, Yaping Zhang, Zora Wu, Haley Li, Wang Jin, Xiaocheng Zhang, Huaxiu Chen, Diyi Yang, Christopher Potts, Christopher D. Manning, and James Zou. Quantifying large language model usage in scientific papers. *Nature Human Behaviour*, 2025.
- [138] Z Liang, W Yu, T Rajpurohit, P Clark, X Zhang, and A Kalyan. Let gpt be a math tutor: Teaching math word problem solvers with customized exercise generation. *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, 2023.

- [139] Zhenwen Liang, Linfeng Song, Yang Li, Tao Yang, Feng Zhang, Haitao Mi, and Dong Yu. Towards solving more challenging imo problems via decoupled reasoning and proving, 2025.
- [140] Zhenwen Liang, Tianyu Yang, Jipeng Zhang, and Xiangliang Zhang. Unimath: A foundational and multimodal mathematical reasoner. *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 7126–7133, 2023.
- [141] Zhenwen Liang, Dian Yu, Xiaoman Pan, Wenlin Yao, Qingkai Zeng, Xiangliang Zhang, and Dong Yu. Mint: Boosting generalization in mathematical reasoning via multi-view fine-tuning. *arXiv preprint arXiv:2307.07951*, 2023.
- [142] Zhenwen Liang, Dian Yu, Wenhao Yu, Wenlin Yao, Zhihan Zhang, Xiangliang Zhang, and Dong Yu. Mathchat: Benchmarking mathematical reasoning and instruction following in multi-turn interactions. *arXiv preprint arXiv:2405.19444*, 2024.
- [143] Haohan Lin, Zhiqing Sun, Sean Welleck, and Yiming Yang. Lean-star: Learning to interleave thinking and proving. In *ICLR*, 2025.
- [144] B Liu, S Bubeck, R Eldan, J Kulkarni, Y Li, A Nguyen, R Ward, and Y Zhang. Augmenting math word problems via iterative question composing. *CoRR*, abs/2312.09241, 2023.
- [145] Chengwu Liu, Ye Yuan, Yichun Yin, Yan Xu, Xin Xu, Zaoyu Chen, Yasheng Wang, Lifeng Shang, Qun Liu, and Ming Zhang. Safe: Enhancing mathematical reasoning in large language models via retrospective step-aware formal verification. In Wanxiang Che, Joyce Nabende, Ekaterina Shutova, and Mohammad Taher Pilehvar, editors, *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 12171–12186, Vienna, Austria, July 2025. Association for Computational Linguistics.
- [146] Hongwei Liu, Zilong Zheng, Yuxuan Qiao, Haodong Duan, Zhiwei Fei, Fengzhe Zhou, Wenwei Zhang, Songyang Zhang, Dahua Lin, and Kai Chen. Mathbench: Evaluating the theory and application proficiency of llms with a hierarchical mathematics benchmark. *arXiv preprint arXiv:2405.12209*, 2024.
- [147] MingShan Liu, Shi Bo, and Jialing Fang. Enhancing mathematical reasoning in large language models with self-consistency-based hallucination detection. *arXiv preprint arXiv:2504.09440*, 2025.
- [148] MingShan Liu, Shi Bo, and Jialing Fang. Enhancing mathematical reasoning in large language models with structured self-consistency. *arXiv preprint*, 2025.
- [149] Mingsheng Liu, Qingming Tang, Jiale Chen, and Steve Renals. Enhancing mathematical reasoning in large language models via preview-based reward models, 2025.
- [150] Wentao Liu, Hanglei Hu, Jie Zhou, Yuyang Ding, Junsong Li, Jiayi Zeng, Mengliang He, Qin Chen, Bo Jiang, Aimin Zhou, et al. Mathematical language models: A survey. *arXiv preprint arXiv:2312.07622*, 2023.
- [151] Zewen Liu. Can large language models adequately perform symbolic reasoning over time series?, 2025.
- [152] Zhengying Liu, Xiaodan Liang, Zhenguo Li, and Haiming Wang. Can llms solve longer math word problems better? *arXiv preprint arXiv:2403.56789*, 2024.
- [153] Zihan Liu, Yang Chen, Mohammad Shoeybi, Bryan Catanzaro, and Wei Ping. Acemath: Advancing frontier math reasoning with post-training and reward modeling. *arXiv preprint arXiv:2412.15084*, 2024. v2, 17 January 2025.
- [154] Zimu Lu, Aojun Zhou, Houxing Ren, Ke Wang, Weikang Shi, Junting Pan, Mingjie Zhan, and Hongsheng Li. Mathgenie: Generating synthetic data with question back-translation for enhancing mathematical reasoning of llms. *arXiv preprint arXiv:2402.16352*, 2024.
- [155] Zimu Lu, Aojun Zhou, Ke Wang, Houxing Ren, Weikang Shi, Junting Pan, Mingjie Zhan, and Hongsheng Li. Mathcoder2: Better math reasoning from continued pretraining on model-translated mathematical code. *Preprint*, arXiv:2410.08196, 2024.
- [156] Haipeng Luo, Qingfeng Sun, Can Xu, Jianguang Lou, Chongyang Tao, et al. RL-theorem prover: Reinforcement learning for large-language-model theorem proving in lean. *arXiv preprint arXiv:2502.06789*, 2025.
- [157] Haipeng Luo, Qingfeng Sun, Can Xu, Pu Zhao, Jianguang Lou, Chongyang Tao, Xiubo Geng, Qingwei Lin, Shifeng Chen, Yansong Tang, and Dongmei Zhang. Wizardmath: Empowering mathematical reasoning for large language models via reinforced evol-instruct. In *Proceedings of the International Conference on Learning Representations (ICLR)*, 2025.
- [158] L Luo, Y Liu, R Liu, S Phatale, H Lara, Y Li, L Shu, J Meng, Y Sun, et al. Improve mathematical reasoning in language models by automated process supervision. *CoRR*, abs/2406.06592, 2024.
- [159] Sina Mahdavi, Wenhao Huang, Lei He, and Kaiyu Yang. Brains vs. bytes: Evaluating olympiad-level proof quality in large language models. *arXiv preprint arXiv:2503.11011*, 2025.
- [160] Qiheng Mao, Rui Du, Junrui Pan, Hongying Liu, Xihe Qiu, Wenchao Ding, Yufei Han, and Weiping Wang. Rulemaker: Error detection and correction for interpretable mathematics in large language models, 2025.
- [161] Yujun Mao, Yoon Kim, and Yilun Zhou. Champ: A competition-level dataset for fine-grained analyses of llms’ mathematical reasoning capabilities. *arXiv preprint arXiv:2401.06961*, 2024.
- [162] Patrick Massot and Jeremy Avigad. Mathematics in lean. *GitHub repository*, 2023.

- [163] John McCarthy, Marvin L Minsky, Nathaniel Rochester, and Claude E Shannon. A proposal for the dartmouth summer research project on artificial intelligence. *AI magazine*, 27(4):12–12, 2006.
- [164] William McCune. Solution of the robbins problem. *Journal of Automated Reasoning*, 19(3):263–276, 1997.
- [165] Iman Mirzadeh, Keivan Alizadeh, Hooman Shahrokhi, Oncel Tuzel, Samy Bengio, and Mehrdad Farajtabar. Gsm-symbolic: Understanding the limitations of mathematical reasoning in large language models. *arXiv preprint arXiv:2410.05229*, 2024.
- [166] Ivan Moshkov et al. Aimo-2 winning solution: Building mathematical reasoning models with the openmathreasoning dataset. *arXiv preprint*, 2025.
- [167] Leonardo de Moura and Sebastian Ullrich. The lean 4 theorem prover and programming language. In *Automated Deduction—CADE 28: 28th International Conference on Automated Deduction, Virtual Event, July 12–15, 2021, Proceedings 28*, pages 625–635. Springer, 2021.
- [168] Logan Murphy, Kaiyu Yang, Jialiang Sun, Zhaoyu Li, Anima Anandkumar, and Xujie Si. Autoformalizing euclidean geometry. *arXiv preprint arXiv:2401.12345*, 2024.
- [169] OpenAI. Gpt-4 technical report. *arXiv preprint arXiv:2303.08774*, 2023.
- [170] OpenAI. OpenAI o4-mini-high Technical Report. *arXiv:2504.09801*, 2025.
- [171] Matthew J. Page, Joanne E. McKenzie, Patrick M. Bossuyt, Isabelle Boutron, and et al. The prisma 2020 statement: An updated guideline for reporting systematic reviews. *BMJ*, 372:n71, 2021.
- [172] Giacomo De Palma, John C. Baez, Christina Lee Yu, Saeed Saremi, Jiantao Jiao, Chunyuan Li, Yuka Nikaido, Igor Razenshteyn, Pascal Vincent, and Greg Yang. Large language models and emergence: A complex systems perspective, 2025.
- [173] K Paster, M Dos Santos, Z Azerbayev, and J Ba. Openwebmath: An open dataset of high-quality mathematical web text. *Proceedings of the 12th International Conference on Learning Representations*, 2024.
- [174] Lawrence C Paulson. Natural deduction as higher-order resolution. *The Journal of logic programming*, 3(3):237–258, 1986.
- [175] Lawrence C Paulson. A preliminary user’s manual for isabelle. Technical report, Computer Laboratory, University of Cambridge, 1988.
- [176] Lawrence C Paulson. Isabelle: The next 700 theorem provers. *Logic and computer science*, pages 361–386, 1990.
- [177] Lawrence C Paulson. Gödel’s incompleteness theorems. <https://isa-afp.org/entries/Incompleteness.html>, 2013. Formal proof development.
- [178] Qizhi Pei et al. Mathfusion: Enhancing mathematic problem-solving of llms through instruction fusion. *arXiv preprint*, 2025.
- [179] Miguel Angel Peñaloza Perez, Bruno Lopez Orozco, Jesus Tadeo Cruz Soto, Michelle Bruno Hernandez, Miguel Angel Alvarado Gonzalez, and Sandra Malagon. Ai4math: A native spanish benchmark for university-level mathematical reasoning in large language models, 2025.
- [180] Gabriel Poesia, David Broman, Nick Haber, and Noah D Goodman. Learning formal mathematics from intrinsic motivation. *arXiv preprint arXiv:2407.00695*, 2024.
- [181] Samyam Rajbhandari, Conglong Li, Zhewei Yao, Minjia Zhang, Reza Yazdani Aminabadi, Ammar Ahmad Awan, Jeff Rasley, and Yuxiong He. Deepspeed-moe: Advancing mixture-of-experts inference and training to power next-generation ai scale. In *Proceedings of the 39th International Conference on Machine Learning (ICML)*, 2022.
- [182] Yanwei Ren, Haotian Zhang, Fuxiang Wu, Jiayan Qiu, Jiaying Huang, Baosheng Yu, and Liu Liu. Sigma: Refining large language model reasoning via sibling-guided monte carlo augmentation, 2025.
- [183] Z.Z. Ren et al. Deepseek-prover-v2: Advancing formal mathematical reasoning via reinforcement learning for subgoal decomposition. *arXiv preprint*, 2025.
- [184] Bernardino Romera-Paredes, Mohammadamin Barekatin, Alexander Novikov, Matej Balog, M Pawan Kumar, Emilien Dupont, Francisco JR Ruiz, Jordan S Ellenberg, Pengming Wang, Omar Fawzi, et al. Mathematical discoveries from program search with large language models. *Nature*, 625(7983):468–476, 2024.
- [185] Bernardino Romera-Paredes, Chris Schmid, Petar Veličković, Karl Tuyls, Andrea Tacchetti, Oriol Vinyals, and Demis Hassabis. Automated discovery of new cap-set constructions with funsearch. *Nature*, 627:125–132, 2024.
- [186] Hassen Saidi, Susmit Jha, and Tuhin Sahai. math-pvs: A large language model framework to map scientific publications to pvs theories. *arXiv preprint arXiv:2310.17064*, 2023.
- [187] Peter Scholze. Half a year of the liquid tensor experiment: Amazing developments. https://www.youtube.com/watch?v=3eg_TKwK5-M, 2022. Accessed: 2023-10-03.
- [188] Sheikh Shafayat, Fahim Tajwar, Ruslan Salakhutdinov, Jeff Schneider, and Andrea Zanette. Can large reasoning models self-train?, 2025.
- [189] Zhihong Shao, Peiyi Wang, Qihao Zhu, Runxin Xu, Junxiao Song, Xiao Bi, Haowei Zhang, Mingchuan Zhang, Yikang Li, Yong Wu, and Daya Guo. Deepseekmath: Pushing the limits of mathematical reasoning in open language models. *arXiv preprint arXiv:2402.03300*, 2024.

- [190] Ziju Shen, Naohao Huang, Fanyi Yang, Yutong Wang, Guoxiong Gao, Tianyi Xu, Jiedong Jiang, Wanyi He, Pu Yang, Mengzhou Sun, Haocheng Ju, Peihao Wu, Bryan Dai, and Bin Dong. Real-prover: Retrieval augmented lean prover for mathematical reasoning, 2025.
- [191] Wenhao Shi, Zhiqiang Hu, Yi Bin, Junhua Liu, Yang Yang, See-Kiong Ng, Lidong Bing, and Roy K.-W. Lee. Math-llava: Bootstrapping mathematical reasoning for multimodal large language models. In *Findings of the Association for Computational Linguistics: EMNLP 2024*, pages 4663–4680, 2024.
- [192] Wenhao Shi, Zhiqiang Hu, Yi Bin, Junhua Liu, Yang Yang, See-Kiong Ng, Lidong Bing, and Roy Ka-Wei Lee. Math-llava: Bootstrapping mathematical reasoning for multimodal large language models. *arXiv preprint arXiv:2406.17294*, 2024.
- [193] Chenglei Si, Weijia Shi, Chen Zhao, Luke Zettlemoyer, and Jordan Boyd-Graber. Getting MoRE out of mixture of language model reasoning experts. In *Findings of the Association for Computational Linguistics: EMNLP 2023*, pages 8234–8249, Singapore, 2023. Association for Computational Linguistics.
- [194] Shiven Sinha, Ameya Prabhu, Ponnurangam Kumaraguru, Siddharth Bhat, and Matthias Bethge. Wu’s method can boost symbolic ai to rival silver medalists and alpageometry to outperform gold medalists at imo geometry. *arXiv preprint arXiv:2404.06405*, 2024.
- [195] John Smith and Michael Johnson. Linc: A neurosymbolic approach for logical reasoning by combining language models with first-order logic provers. *arXiv preprint arXiv:2402.12345*, 2024.
- [196] Filip Smola and Jacques D Fleuriot. Hyperdual numbers and forward differentiation. <https://isa-afp.org/entries/Hyperdual.html>, 2021. Formal proof development.
- [197] Peiyang Song, Alex Gu, Kaiyu Yang, and Rahul Chalamala. Fvel: Interactive formal verification environment with large language models via theorem proving. *arXiv preprint arXiv:2402.03456*, 2024.
- [198] Peiyang Song, Kaiyu Yang, and Anima Anandkumar. Towards large language models as copilots for theorem proving in lean. *arXiv preprint arXiv:2404.12534*, 2024.
- [199] Z Sprague, F Yin, J D Rodriguez, D Jiang, M Wadhwa, P Singhal, X Zhao, X Ye, K Mahowald, and G Durrett. To cot or not to cot? chain-of-thought helps mainly on math and symbolic reasoning. *CoRR*, abs/2409.12183, 2024a.
- [200] Gaurav Srivastava, Aafiya Hussain, Sriram Srinivasan, and Xuan Wang. Llmthinkbench: Towards basic math reasoning and overthinking in large language models, 2025.
- [201] P Srivastava, M Malik, T Ganu, and D Roth. Evaluating llms’ mathematical reasoning in financial document question answering. *Findings of the Association for Computational Linguistics*, 2024.
- [202] Saksham Sahai Srivastava and Ashutosh Gandhi. Mathdivide: Improved mathematical reasoning by large language models. *arXiv preprint arXiv:2405.13004*, 2024.
- [203] Anselm R. Strohmaier, Wim Van Dooren, Kathrin Seßler, Brian Greer, and Lieven Verschaffel. Large language models don’t make sense of word problems: A study on mathematical reasoning, 2025.
- [204] Haoxiang Sun, Yingqian Min, Zhipeng Chen, Wayne Xin Zhao, Zheng Liu, Zhongyuan Wang, Lei Fang, and Ji-Rong Wen. Challenging the boundaries of reasoning: An olympiad-level math benchmark for large language models, 2025.
- [205] Yuhong Sun, Zhangyue Yin, Qipeng Guo, Jiawen Wu, Xipeng Qiu, and Hui Zhao. Benchmarking hallucination in large language models based on unanswerable math word problems. *arXiv preprint arXiv:2403.03558*, 2024.
- [206] Yuhong Sun, Zhangyue Yin, Qipeng Guo, Jiawen Wu, Xipeng Qiu, and Hui Zhao. Mathsensei: A tool-augmented large language model for mathematical reasoning. *arXiv preprint arXiv:2407.01234*, 2024.
- [207] Christian Szegedy. A promising path towards autoformalization and general artificial intelligence. In *International Conference on Intelligent Computer Mathematics*, pages 3–20. Springer, 2020.
- [208] Suraiya Tairin, Shohaib Mahmud, Haiying Shen, and Anand Iyer. emoe: Task-aware memory efficient mixture-of-experts-based model inference. *arXiv preprint*, arXiv:2503.06823, 2025.
- [209] Zhengyang Tang, Xingxing Zhang, Benyou Wang, and Furu Wei. Mathscale: Scaling instruction tuning for mathematical reasoning. In *Proceedings of the 41st International Conference on Machine Learning (ICML)*, 2024.
- [210] The mathlib Community. The lean mathematical library, 2020. CPP ’20.
- [211] S Toshniwal, W Du, I Moshkov, B Kisananin, A Ayrapetyan, and I Gitman. Openmathinstruct-2: Accelerating ai for math with massive open-source instruction data. *CoRR*, abs/2410.01560, 2024a.
- [212] S Toshniwal, I Moshkov, S Narenthiran, D Gitman, F Jia, and I Gitman. Openmathinstruct-1: A 1.8 million math instruction tuning dataset. *Advances in Neural Information Processing Systems* 38, 2024b.
- [213] Hugo Touvron, Thibaut Lavril, Gautier Izacard, Xavier Martinet, Marie-Anne Lachaux, Timothée Lacroix, Baptiste Rozière, Naman Goyal, Eric Hambro, Azhar Faisal, et al. Llama: Open and efficient foundation language models. *arXiv preprint arXiv:2302.13971*, 2023.
- [214] Trieu H. Trinh, Yuhuai Wu, Quoc V. Le, He He, et al. Solving olympiad geometry without human demonstrations. *Nature*, 625:476–482, 2024.
- [215] Haiming Wang, Zhengying Liu, Xiaodan Liang, and Zhenguo Li. Branch-train-mix: Mixing expert llms into a mixture-of-experts llm. *arXiv preprint arXiv:2404.01234*, 2024.

- [216] Haiming Wang, Zhengying Liu, Xiaodan Liang, and Zhenguo Li. Mumath: Multi-perspective data augmentation for mathematical reasoning in large language models. *arXiv preprint arXiv:2406.03548*, 2024.
- [217] Jiayu Wang, Yifei Ming, Zixuan Ke, Caiming Xiong, Shafiq Joty, Aws Albarghouthi, and Frederic Sala. Beyond accuracy: Dissecting mathematical reasoning for llms, 2025.
- [218] Ke Wang, Junting Pan, Weikang Shi, Zimu Lu, Houxing Ren, Aojun Zhou, Mingjie Zhan, and Hongsheng Li. Measuring multimodal mathematical reasoning with math-vision dataset. In *Proceedings of the 37th Conference on Neural Information Processing Systems Datasets and Benchmarks Track (NeurIPS)*, 2024.
- [219] Liang Wang, Isabel Fischer, and Karim Ibrahim. Agentic hypothesis generation with large language models: A survey. In *ICLR 2025 Workshop on Scientific Discovery with AI*, 2025.
- [220] Peiyi Wang, Lei Li, Zhihong Shao, Runxin Xu, Damai Dai, Yifei Li, Deli Chen, Yu Wu, and Zhifang Sui. Math-shepherd: Verify and reinforce llms step-by-step without human annotations. *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 9426–9439, 2024.
- [221] Qingxiang Wang, Cezary Kaliszyk, and Josef Urban. First experiments with neural translation of informal to formal mathematics. In *Intelligent Computer Mathematics: 11th International Conference, CICM 2018, Hagenberg, Austria, August 13-17, 2018, Proceedings 11*, pages 255–270. Springer, 2018.
- [222] Ruida Wang, Yuxin Li, Yi R. Fung, and Tong Zhang. Let’s reason formally: Natural-formal hybrid reasoning enhances llm’s math capability, 2025.
- [223] Siyun Wang, Yibo Miao, Hao Li, Tianyu Chang, Kaixuan Chen, Lei Xia, Songyang Gao, Jichuan Tian, Yuhang Lu, Longtao Guo, Dong Li, Xiang Ao, Min Yang, Rong Pan, and Feida Zhu. Can llms understand math? exploring the pitfalls in mathematical reasoning of large language models, 2025.
- [224] Tan Wang, Derui Qiu, Qinglong Xia, Chongyu Luo, Bin Qi, Sihao Hu, Huan Li, Qi Liu, and Jun Zhou. A survey on large language models for mathematical reasoning, 2025.
- [225] Yiming Wang, Pei Zhang, Jialong Tang, Haoran Wei, Baosong Yang, Rui Wang, Chenshu Sun, Feitong Sun, Jiran Zhang, Junxuan Wu, Qiqian Cang, Yichang Zhang, Fei Huang, Junyang Lin, and Jingren Zhou. Polymath: Evaluating mathematical reasoning in multilingual contexts, 2025.
- [226] Zhongtian Wang, Yuzhe Tian, Guangwei Chen, Wei Chen, Jing Lin, and Jingyang Xiao. A survey of mathematical reasoning in the era of multimodal large language model: Benchmark, method & challenges. In Wanxiang Che, Joyce Nabende, Ekaterina Shutova, and Mohammad Taher Pilehvar, editors, *Findings of the Association for Computational Linguistics: ACL 2025*, pages 12705–12716, Vienna, Austria, July 2025. Association for Computational Linguistics.
- [227] Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Fei Xia, Ed Chi, Quoc V Le, and Denny Zhou. Chain-of-thought prompting elicits reasoning in large language models. *Advances in Neural Information Processing Systems*, 35:24824–24837, 2022.
- [228] T Wei, J Luan, W Liu, S Dong, and B Wang. Cmath: can your language model pass chinese elementary school math test? *CoRR*, abs/2306.16636, 2023.
- [229] Sean Welleck, Jiacheng Liu, Ximing Lu, Hannaneh Hajishirzi, and Yejin Choi. Naturalprover: Grounded mathematical proof generation with language models. *Advances in Neural Information Processing Systems*, 35:4913–4927, 2022.
- [230] Ke Weng, Lun Du, Sirui Li, Wangyue Lu, Haozhe Sun, Hengyu Liu, and Tiancheng Zhang. Autoformalization in the era of large language models: A survey, 2025.
- [231] Freek Wiedijk. The “de bruijn factor”. <https://www.cs.ru.nl/~freek/100/>, 2012. Accessed: 2023-09-28.
- [232] Catherine Williams, David Green, and Michael Smith. Revisiting smoe language models by evaluating inefficiencies with task-specific expert pruning. *arXiv preprint arXiv:2404.12345*, 2024.
- [233] Catherine Williams, Daniel Smith, and Robert Carter. Decomposing the enigma: Subgoal-based demonstration learning for formal theorem proving. *arXiv preprint arXiv:2401.67890*, 2024.
- [234] Haoyuan Wu, Haisheng Zheng, Zhuolun He, and Bei Yu. Parameter-efficient sparsity crafting from dense to mixture-of-experts for instruction tuning on general tasks. *arXiv preprint arXiv:2401.02731*, 2024.
- [235] Minchao Wu, Michael Norrish, Christian Walder, and Amir Dezfouli. Tacticzero: Learning to prove theorems from scratch with deep reinforcement learning. In *Advances in Neural Information Processing Systems*, pages 9330–9342, 2021.
- [236] Yuhuai Wu, Albert Qiaoju Jiang, Wenda Li, Markus Rabe, Charles Staats, Mateja Jamnik, and Christian Szegedy. Autoformalization with large language models. In *Advances in Neural Information Processing Systems*, pages 32353–32368, 2022.
- [237] Yutong Wu, Di Huang, Ruosi Wan, Yue Peng, Shijie Shang, Chenrui Cao, Lei Qi, Rui Zhang, Zidong Du, Jie Yan, and Xing Hu. Stepfun-formalizer: Unlocking the autoformalization potential of llms through knowledge-reasoning fusion, 2025.
- [238] xAI Research. Grok 3 Technical Report. <https://x.ai/research/grok3>, 2025. Version 1.0, May 2025.
- [239] Huajian Xin, Daya Guo, Zhihong Shao, ZZ Ren, Qihao Zhu, Bo Liu, Chong Ruan, Wenda Li, and Xiaodan Liang. Deepseek-prover: Advancing theorem proving in llms through large-scale synthetic data. In *International Conference*

on Learning Representations, 2024.

- [240] Huajian Xin, Zhihong Shao, Wenda Li, and Daya Guo. Deepseek-prover v1.5: Reinforcement learning and tree search for formal proof synthesis. *arXiv preprint arXiv:2411.09876*, 2024.
- [241] Huajian Xin, Haiming Wang, Chuanyang Zheng, Lin Li, Zhengying Liu, et al. Lego-prover: Neural theorem proving with growing libraries. *arXiv preprint arXiv:2310.00656*, 2023.
- [242] Tairan Xu, Leyang Xue, Zhan Lu, Adrian Jackson, and Luo Mai. Moe-gen: High-throughput moe inference on a single gpu with module-based batching. *arXiv preprint*, arXiv:2503.09716, 2025.
- [243] Xin Xu et al. Ugmathbench: A diverse and dynamic benchmark for undergraduate-level mathematical reasoning. *arXiv preprint*, 2025.
- [244] Boyang Xue, Qi Zhu, Rui Wang, Sheng Wang, Hongru Wang, Fei Mi, Yasheng Wang, Lifeng Shang, Qun Liu, and Kam-Fai Wong. Reliablemath: Benchmark of reliable mathematical reasoning on large language models, 2025.
- [245] Y Yan, J Su, J He, F Fu, X Zheng, K Lyu, S Wang, Q Wang, H Wen, and X Hu. A survey of mathematical reasoning in the era of multimodal large language model: Benchmark, method challenges. *CoRR*, abs/2412.11936, 2024.
- [246] Yibo Yan, Shen Wang, Jiahao Huo, Hang Li, Boyan Li, Jiamin Su, Xiong Gao, Yi-Fan Zhang, Tianlong Xu, Zhendong Chu, et al. Errorradar: Benchmarking complex mathematical reasoning of multimodal large language models via error detection. *arXiv preprint arXiv:2410.04509*, 2024.
- [247] Yibo Yan, Shen Wang, Jiahao Huo, Philip S. Yu, Xuming Hu, and Qingsong Wen. Mathagent: A multi-expert planner for proof construction and conjecture suggestion. *arXiv preprint arXiv:2503.18132*, 2025.
- [248] Yibo Yan, Shen Wang, Jiahao Huo, Philip S Yu, Xuming Hu, and Qingsong Wen. Mathagent: Leveraging a mixture-of-math-agent framework for real-world multimodal mathematical error detection. *arXiv preprint arXiv:2503.18132*, 2025.
- [249] An Yang, Beichen Zhang, Binyuan Hui, Bofei Gao, Bowen Yu, Chengpeng Li, Dayiheng Liu, Jianhong Tu, Jingren Zhou, Junyang Lin, et al. Qwen2.5-math technical report: Toward mathematical expert model via self-improvement. *arXiv preprint arXiv:2409.12122*, 2024.
- [250] Kaiyu Yang, Albert Q. Jiang, Mateja Jamnik, Jason Rute, and Yuhuai Wu. Mathbench: Evaluating the theory and application proficiency of llms with a hierarchical mathematics benchmark. *arXiv preprint arXiv:2404.56789*, 2024.
- [251] Kaiyu Yang, Logan Murphy, Zhaozhong Liu, Jialiang Sun, and Mateja Jamnik. Steamroller problems: An evaluation of llm reasoning capability with automated theorem prover strategies. *arXiv preprint arXiv:2311.45678*, 2023.
- [252] Kaiyu Yang, Aidan Swope, Alex Gu, Peiyang Song, and Saad Godil. Subgoalxl: Subgoal-based expert learning for theorem proving. *arXiv preprint arXiv:2401.01234*, 2023.
- [253] Kaiyu Yang, Aidan M. Swope, Alex Gu, Rahul Chalamala, Peiyang Song, Shixing Yu, Saad Godil, Ryan Prenger, and Anima Anandkumar. Leandojo: Theorem proving with retrieval-augmented language models. *arXiv preprint arXiv:2310.08488*, 2023.
- [254] Wenkai Yang, Jingwen Chen, Yankai Lin, and Ji-Rong Wen. Deepcritic: Deliberate critique with large language models, 2025.
- [255] Yi Yang, Guang Yang, Fengzhen Lin, Shuchang Liu, and Zhenyu Hou. Who reasons in the large language models?, 2025.
- [256] Zhaokun Yang, Xinhai Chen, Yufei Geng, Yu Zhang, Bo Liu, Che Liu, Hang Dong, and Gang Zhang. Warriormath: Enhancing the mathematical ability of large language models through self-improvement, 2025.
- [257] Zhen Yang, Jinhao Chen, Zhengxiao Du, Wenmeng Yu, Weihang Wang, Wenyi Hong, Zhihuan Jiang, Bin Xu, Yuxiao Dong, and Jie Tang. Mathglm-vision: Solving mathematical problems with multi-modal large language model. *arXiv preprint arXiv:2409.13729*, 2024.
- [258] Zhen Yang, Ming Ding, Qingsong Lv, Zhihuan Jiang, Zehai He, Yuyi Guo, Jinfeng Bai, and Jie Tang. Gpt can solve mathematical problems without a calculator. *arXiv preprint arXiv:2309.03241*, 2023.
- [259] Zhen Yang and et al. Qwen2.5-math technical report: Toward mathematical expert model via self-improvement. *arXiv preprint arXiv:2409.12122*, 2024.
- [260] Jian Yao, Ran Cheng, and Kay Chen Tan. Var-math: Probing true mathematical reasoning in large language models via symbolic multi-instance benchmarks, 2025.
- [261] T Ye, Z Xu, Y Li, and Z Allen-Zhu. Physics of language models: Part 2.2, how to learn from mistakes on grade-school math problems. *CoRR*, abs/2408.16293, 2024.
- [262] Huaiyuan Ying, Zijian Wu, Yihan Geng, Jiayu Wang, Dahua Lin, and Kai Chen. Lean workbook: A large-scale lean problem set formalized from natural language math problems. *arXiv preprint arXiv:2406.03847*, 2024.
- [263] Huaiyuan Ying, Shuo Zhang, Linyang Li, Zhejian Zhou, Yunfan Shao, Zhaoye Fei, Yichuan Ma, Jiawei Hong, Kuikun Liu, Ziyi Wang, Yudong Wang, Zijian Wu, Shuaibin Li, Fengzhe Zhou, Hongwei Liu, Songyang Zhang, Wenwei Zhang, Hang Yan, Xipeng Qiu, Jiayu Wang, Kai Chen, and Dahua Lin. Internlm-math: Open math large language models toward verifiable reasoning, 2024.

- [264] Hiroshi Yoshihara, Taiki Yamaguchi, and Yuichi Inoue. A practical two-stage recipe for mathematical llms: Maximizing accuracy with sft and efficiency with reinforcement learning, 2025. ICML 2025 Workshop: The Second AI for MATH.
- [265] Longhui Yu, Weisen Jiang, Han Shi, Jincheng Yu, Zhengying Liu, Yu Zhang, James T. Kwok, Zhenguo Li, Adrian Weller, and Weiyang Liu. Metamath: Bootstrap your own mathematical questions for large language models. In *Proceedings of the International Conference on Learning Representations (ICLR)*, 2024.
- [266] Yiyao Yu, Yuxiang Zhang, Dongdong Zhang, Xiao Liang, Hengyuan Zhang, Xingxing Zhang, Mahmoud Khademi, Hany Hassan Awadalla, Junjie Wang, Yujiu Yang, and Furu Wei. Chain-of-reasoning: Towards unified mathematical reasoning in large language models via a multi-paradigm perspective. In Wanxiang Che, Joyce Nabende, Ekaterina Shutova, and Mohammad Taher Pilehvar, editors, *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 24914–24937, Vienna, Austria, July 2025. Association for Computational Linguistics.
- [267] Zhouliang Yu, Ruotian Peng, Haozhan Cao, Tianduo Wang, Jingtao Wang, Qiyang Sun, Hang Song, Jianqiao Li, Yatong Zhou, Yuenan Zhou, Xuanfei Zhu, Haowen Tang, Zhaotian Xie, Ke Zhang, Xisheng Dai, Yunfeng Cai, Lianghua He, Lanqing Hong, Zheng Qin, Yingming Li, and Junjun Huang. Formalmath: Benchmarking formal mathematical reasoning of large language models, 2025.
- [268] Xiang Yue, Xingwei Qu, Ge Zhang, Yao Fu, Wen-hao Huang, Huan Sun, Yu Su, and Wenhui Chen. Mammoth: Building math generalist models through hybrid instruction tuning. *arXiv preprint arXiv:2309.05653*, 2023.
- [269] Xiang Yue, Tuney Zheng, Ge Zhang, and Wenhui Chen. Mammoth2: Scaling instructions from the web. *arXiv preprint arXiv:2405.03548*, 2024.
- [270] Lei Zan, Keli Zhang, Ruichu Cai, and Lujia Pan. Cama: Enhancing mathematical reasoning in large language models with causal knowledge, 2025.
- [271] Hanqing Zeng, Yinglong Xia, Zhuokai Zhao, Gilbert Jiang, Qiang Zhang, Jiayi Liu, Lizhu Zhang, Xiangjun Fan, and Benyu Zhang. S’more: Structural mixture of residual experts for llm fine-tuning. *arXiv preprint arXiv:2504.06426*, 2025. v1, April 2025.
- [272] Liang Zeng and Liangjun Zhong. Skywork-math: Data scaling laws for mathematical reasoning in llms – the story goes on. In *Proceedings of the NeurIPS 2024 MATH-AI Workshop*, 2024.
- [273] Liang Zeng, Liangjun Zhong, Liang Zhao, Tianwen Wei, Liu Yang, Jujie He, Cheng Cheng, Rui Hu, Yang Liu, Shuicheng Yan, et al. Skywork-math: Data scaling laws for mathematical reasoning in large language models—the story goes on. *arXiv preprint arXiv:2407.08348*, 2024.
- [274] Boning Zhang, Chengxi Li, and Kai Fan. Marioeval: Evaluate your math llm with your math llm – a mathematical dataset evaluation toolkit. *arXiv preprint arXiv:2404.13925*, 2024.
- [275] Boning Zhang, Chengxi Li, and Kai Fan. Mathodyssey: Benchmarking mathematical problem-solving skills in large language models using odyssey math data. *arXiv preprint arXiv:2404.13925*, 2024.
- [276] Chenyu Zhang, Wenqi Li, Meiting Wang, Yuan Zhao, and Ming Zhou. Mathverse: A comprehensive benchmark for visual mathematical reasoning. *arXiv preprint arXiv:2406.12345*, 2024.
- [277] Jianwei Zhang, Marco Valentino, and André Freitas. Formalising complex mathematical definitions with large language models. *arXiv preprint arXiv:2501.01234*, 2025.
- [278] Jiaxin Zhang, Zhongzhi Li, Mingliang Zhang, Fei Yin, Chenglin Liu, and Yashar Moshfeghi. Geoeval: benchmark for evaluating llms and multi-modal models on geometry problem-solving. *arXiv preprint arXiv:2402.10104*, 2024.
- [279] Jie Zhang, Cezara Petriuc, Kristina Nikolić, and Florian Tramèr. Realmath: A continuous benchmark for evaluating language models on research-level mathematics, 2025.
- [280] Xiang Zhang, Junqiao Zhao, Huadeng Zhu, Shizhu He, and Kang Liu. A benchmark for evaluating mathematical creativity of large language models, 2025.
- [281] Yuxiang Zhang, Lei Qi, and Zhiyuan Liu. Consistent autoformalization via retrieval-augmented correction loops. *arXiv preprint arXiv:2410.07654*, 2024.
- [282] Zeren Zhang, Jo-Ku Cheng, Jingyang Deng, Lu Tian, Jinwen Ma, Ziran Qin, Xiaokai Zhang, Na Zhu, and Tuo Leng. Diagram formalization enhanced multi-modal geometry problem solver. *arXiv preprint arXiv:2409.04214*, 2024.
- [283] Zichen Zhang, Jipeng Zhang, Albert Qiaolu He, Honghua Zhang, and Xifeng Yan. Leanconjecturer: Automatic generation of mathematical conjectures for theorem proving, 2025.
- [284] Ziyin Zhang, Jiahao Xu, Zhiwei He, Tian Liang, Qiuzhi Liu, Yansi Li, Linfeng Song, Zhenwen Liang, Zhuosheng Zhang, Rui Wang, Zhaopeng Tu, Haitao Mi, and Dong Yu. Deeptheorem: Advancing llm reasoning for theorem proving through natural language and reinforcement learning, 2025.
- [285] Wayne Xin Zhao, Kun Zhou, Junyi Li, Tianyi Tang, Xiaolei Wang, Yupeng Hou, Yingqian Min, Beichen Zhang, Junjie Zhang, Zican Dong, et al. A survey of large language models. *arXiv preprint arXiv:2303.18223*, 2023.
- [286] Xueliang Zhao, Wei Wu, Jian Guan, and Lingpeng Kong. Promptcot: Synthesizing olympiad-level problems for mathematical reasoning in large language models. In Wanxiang Che, Joyce Nabende, Ekaterina Shutova, and Mohammad Taher Pilehvar, editors, *Findings of the Association for Computational Linguistics: ACL 2025*, pages

- 18167–18188, Vienna, Austria, July 2025. Association for Computational Linguistics.
- [287] Zikun Zhao, Linyang Li, Zihao Chen, Jidong Tu, Baoshan Kan, Qipeng Guo, and Xipeng Qiu. Revisiting reinforcement learning for llm reasoning from a cross-domain perspective, 2025.
 - [288] Q Zhong, K Wang, Z Xu, J Liu, L Ding, M Farajtabar, Y Tao, and D Shen. Achieving >97 CoRR, abs/2404.14963, 2024.
 - [289] Kun Zhou, Beichen Zhang, Jiapeng Wang, Wayne X Chen, Zhipeng Zhao, Jing Sha, Zhichao Sheng, Qian Wang, and Ji-Rong Wen. Jiuzhang 3.0: Efficiently improving mathematical reasoning by training small data synthesis models. *arXiv preprint arXiv:2405.14365*, 2024.
 - [290] Yichi Zhou, Jianqiu Zhao, Yongxin Zhang, Bohan Wang, Siran Wang, Luoxin Chen, Jiahui Wang, Haowei Chen, Allan Jie, Xinbo Zhang, Haocheng Wang, Luong Trung, Rong Ye, Phan Nhat Hoang, Huishuai Zhang, Peng Sun, and Hang Li. Solving formal math problems by decomposition and iterative reflection, 2025.
 - [291] Wei Zhuang, Lin Tang, Ziyi Chen, and Yixin Xu. Math-puma: Progressive upward multimodal alignment for diagram-aware mathematical reasoning. *AAAI Conference on Artificial Intelligence*, 2025.
 - [292] Wenwen Zhuang, Xin Huang, Xiantao Zhang, and Jin Zeng. Math-puma: Progressive upward multimodal alignment to enhance mathematical reasoning. *arXiv preprint arXiv:2408.08640*, 2024.
 - [293] —. Advancing multi-step mathematical reasoning in large language models through multi-layered self-reflection with auto-prompting, 2025. Author names to be updated; see arXiv record.